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Osteoporosis Risk Prediction

A Dataset for Exploring Lifestyle Factors and Osteoporosis Risk

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1. Real Problem Presentation

1.1 The Silent Nature of Osteoporosis

Osteoporosis represents a unique challenge in modern healthcare due to its asymptomatic progression until catastrophic events occur. Unlike acute conditions that prompt immediate medical attention through obvious symptoms, osteoporosis earned its moniker as the "silent disease" due to its stealthy progression. Patients often remain unaware of their condition until a sudden fracture reveals the extensive bone deterioration that has occurred over years or decades.

The disease's silent progression creates a critical window where intervention could prevent severe outcomes so it is important to recognize it as early as possible because:

The disease's silent nature makes it particularly dangerous for several reasons:

- Bone deterioration occurs gradually without noticeable symptoms.
- The first sign is often a sudden fracture during routine activities.
- By the time of diagnosis, significant bone loss has typically already occurred.
- The condition can affect any bone in the body, such as neck, skull, chest, foot, wrist and etc and making its impact unpredictable and widespread.

1.2 Comprehensive Impact Assessment

Physical Impact

- Increased risk of fractures in various body parts:
 - Hip fractures (associated with 20-24% mortality rate within first year).
 - Vertebral fractures leading to height loss and chronic pain.
 - Wrist fractures affecting daily activities and independence
- Reduced mobility and physical capability
- Chronic pain management challenges
- Postural changes affecting body mechanics

The chart below displays the number of deaths related to LBMD (Low Bone Mineral Density) in 2019.

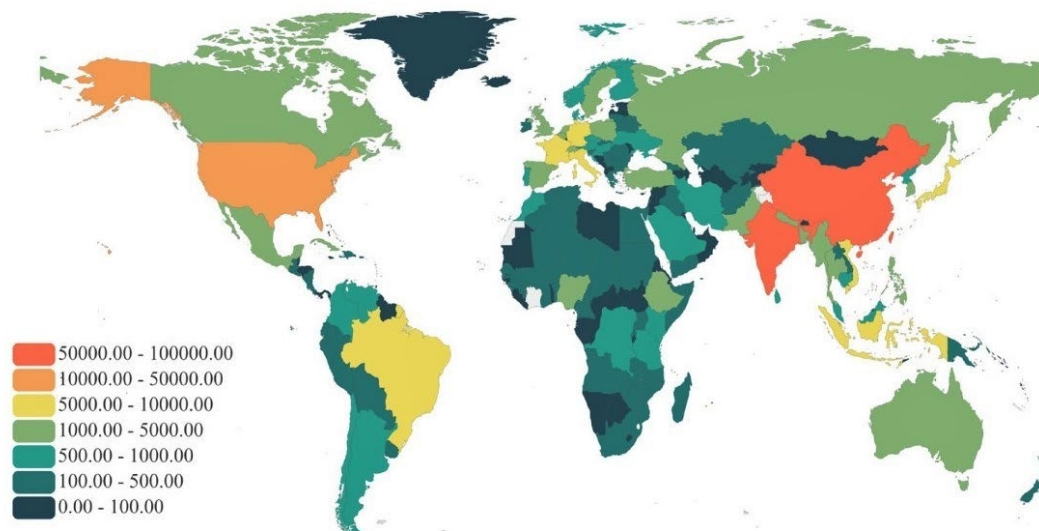


Figure1 : Global map of death number related to LBMD in 2019

Osteoporosis, beyond its physical effects on the body, can significantly impact various aspects of life, presenting both personal and societal challenges.

Quality of Life Impact:

- **Decreased independence in daily activities:** Individuals with osteoporosis may struggle with routine tasks such as dressing, cooking, or cleaning due to pain or limited mobility.
- **Reduced social participation:** Fear of fractures and falls often leads people with osteoporosis to avoid social engagements, reducing their connection to family and friends.
- **Increased fear of falling:** The heightened concern about falling may result in activity limitations, further decreasing mobility and quality of life.
- **Sleep disturbances:** Persistent pain or discomfort from fractures or bone weakness can interfere with rest, leading to insomnia or poor-quality sleep.
- **Mental health challenges:** The limitations caused by osteoporosis can lead to feelings of isolation, anxiety, and even depression, negatively affecting emotional well-being.

Economic Impact:

- Direct medical costs: Treating fractures, hospitalizations, and ongoing medical care for osteoporosis-related issues can be expensive.
- Long-term rehabilitation expenses: Patients may require physical therapy and long-term rehabilitation to regain mobility or manage pain after fractures.
- Lost productivity and work capacity: Due to physical limitations or frequent medical appointments, individuals with osteoporosis may experience reduced work capacity or take time off work, impacting income and economic stability.
- Caregiver burden: Family members or hired caregivers often face the emotional and financial burden of providing support, leading to additional costs.
- Preventive medication and monitoring expenses: Ongoing treatment, medications, and regular medical check-ups are necessary to manage osteoporosis and prevent further bone loss or fractures, adding to the long-term financial impact.

Osteoporosis, therefore, poses a significant challenge not only to physical health but also to mental, social, and economic well-being.

2. Data Understanding

In this section, we are going to explore a specific dataset titled “*A Dataset for Exploring Lifestyle Factors and Osteoporosis Risk*” available on the Kaggle website.

2.1 Dataset Overview

The dataset provides lifestyle details of individuals with and without osteoporosis. Our analysis utilizes a comprehensive dataset of 1,958 individuals, capturing multiple dimensions:

- Demographic factors (age, gender, ethnicity)
- Physiological markers (hormonal status, body weight)
- Lifestyle factors (physical activity, smoking, alcohol consumption)
- Medical history (prior fractures, existing conditions, family history, medication use)
- Nutritional factors (calcium and vitamin D intake)

2.2 Dataset Variables: Detailed Definition

Target Variable:

The target variable in this dataset is "Osteoporosis Status," which represents whether an individual has been diagnosed with osteoporosis.

➤ Variable Type: Binary Categorical

➤ Possible Values: $\begin{cases} 0 = \text{No osteoporosis present} \\ 1 = \text{Osteoporosis present} \end{cases}$

Independent Variables:

An independent variable is a factor, characteristic, or attribute that can potentially influence or predict the outcome. In research and statistical analysis, these are the variables that are manipulated or measured to understand their impact on the dependent variable (in this case, osteoporosis status).

As we explain in the previous section, our independent variables fall into 5 groups, demographic factors, physiological markers, lifestyle factors, medical history and nutritional factors.

In another perspective we can categorize them based on their types, that if they are categorical or binary or numeric.

In general, we have:

- 10 nominal binary variables.
- 2 nominal variables.
- 2 numerical variables. (discrete)

Although we can consider some of the binary variables as nominal, since their possible values describe two distinct conditions, for example for the medication use, it can be either None or Corticosteroids (if someone use Corticosteroid drugs or not), we assume it as binary to facilitate our analysis without losing information.

Because we can typically consider their result as 0 or 1. For instance, zero if no medication is used and 1 if the person consumes Corticosteroid.

In the next chart we cluster the variables, based on their types, and shows the possible values for each of them.

Variable Type	Variable Name	Possible Values
Numerical	Age	Measured in years (integer)
	Id	Unique identifier
Binary	Gender	Male / Female
	Hormonal Changes	Normal / Postmenopausal
	Family History	Yes / No
	Body Weight	Normal / Underweight
	Calcium Intake	Low / Adequate
	Vitamin D Intake	Insufficient / Sufficient
	Physical Activity	Sedentary / Active
	Smoking	Yes / No
	Alcohol Consumption	None / Moderate
	Medications Use	None / Corticosteroids
	Prior Fractures	Yes / No
Nominal	Race / Ethnicity	Caucasian, African American, Asian, etc.
	Medical Conditions	None, Rheumatoid Arthritis, Hyperthyroidism

3. Data Preparation

In this section, we focus on preparing the dataset to ensure it is clean, consistent, and ready for analysis. Data preparation is crucial to building a robust foundation for exploring relationships between variables and developing accurate models.

3.1 Data Cleaning

In the following we will do some steps for cleaning our data.

✓ **Check for missing values:**

The dataset consists of 1,958 entries across 16 columns, and all columns are complete with no missing values. Therefore, there is no need to identify missing data patterns or implement imputation strategies, which are typically critical steps in handling datasets with incomplete information.

It is also worth nothing to mention that in our dataset, some categorical variables have a value of "None," which represents the absence of a specific category. However, Python often interprets "None" as a missing value (NaN), which can lead to it being ignored in visualizations or analyses. To ensure these values are treated as a distinct category rather than missing data, we replace "None" with "Nothing."

✓ **Inspect and remove duplicates:**

In our dataset there are no duplicate rows based on the *Id* column.

✓ **Removing unnecessary data:**

In this section, we identify and remove columns that do not contribute meaningful information to the analysis. The *Id* column is a numerical identifier used solely to check for duplicate entries. Since it does not provide any analytical value or insights about osteoporosis, we omit it from the dataset after confirming there are no duplicate records.

✓ **Data validation:**

In this step, we validate the dataset to ensure its integrity and consistency. This includes checking for typos or inconsistencies in categorical variables and verifying that all binary variables, particularly the target variable 'Osteoporosis,' are correctly encoded as binary values (0/1).

✓ **Illogical inconsistencies:**

Addressing data inconsistencies is an essential part of data cleaning. In this process, we identify and correct logical errors that contradict known scientific or medical principles. In this dataset, we identified and removed rows with the following inconsistencies:

1. Female participants younger than 30 years of age classified as postmenopausal represent a logical inconsistency, Menopause typically occurs between the ages of 45 and 55 (with an average age of 51). It is extremely rare for it to happen before the age of 30, and such classifications are highly unlikely according to medical understanding.¹
2. Although menopause is a biological impossibility for men, males can experience age-related hormonal changes similar to the concept of “postmenopause”, such as testosterone reduction or late-onset hypogonadism. However, these changes typically occur later in life, usually in their 50s or beyond. Therefore, classifying males under 40 as postmenopausal is completely inconsistent with current medical understanding, as such hormonal changes do not usually manifest at that age.²
3. Age values cannot be negative. These types of errors often arise from data entry mistakes and must be resolved.

¹ Based on NHS (National Health Service), premature menopause is estimated to affect 1% of women under the age of 40 years and 0.1% of women under the age of 30 years.

² Based on NHS (National Health Service), <https://www.nhs.uk/conditions/male-menopause/>

✓ **Handle outliers:**

Outliers in the dataset can distort statistical analyses and affect the reliability of results. Since the only numeric variable in this dataset is *Age*, we define a realistic age range of 10 to 90 and flag any values outside this range for correction.

3.2 Encoding Categorical Binary Variables

In this section, we encoded all binary categorical variables into numerical form using the “LabelEncoder” from the “sklearn” library in Python. Binary encoding was chosen for its simplicity and compatibility with models, making the dataset easier to interpret and analyze without introducing additional complexity. Each variable was assigned a value of 0 for one category and 1 for the other. For instance, the 'Gender' variable was encoded as 0 for 'Female' and 1 for 'Male'. Actually, this library performs this process alphabetically, starting from 0. Additionally, the 'Race/Ethnicity' and 'Medical Condition' variables were also encoded. This approach ensures the data is ready for statistical analysis and machine learning algorithms, which require numerical inputs.

The tables below illustrate the encoding schemes applied to both binary categorical variables and nominal categorical variables in the dataset.

Variable Name	0	1	2
Medical Conditions	Hyperthyroidism	None	Rheumatoid Arthritis
Race/Ethnicity	African American	Asian	Caucasian

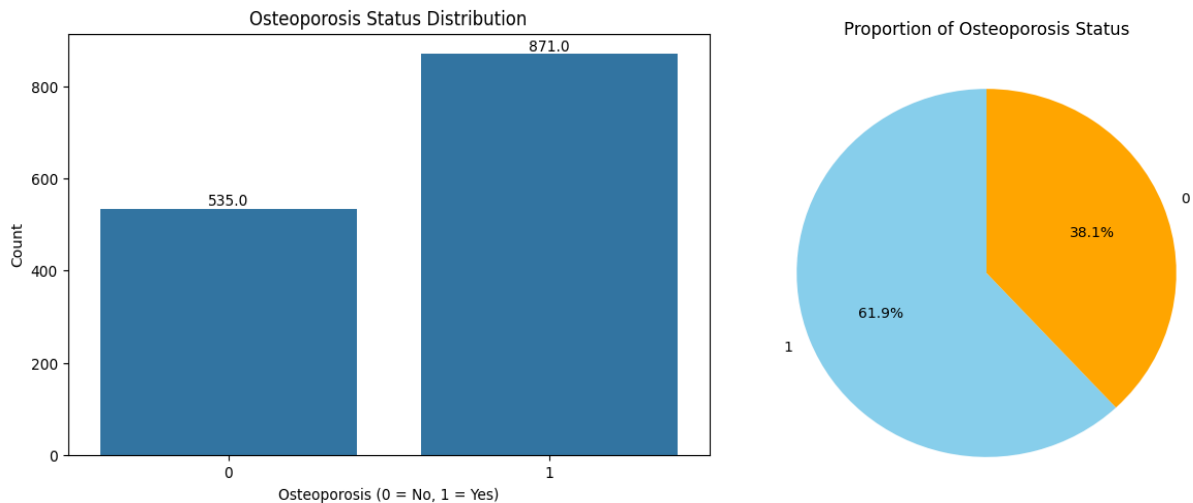
Variable Name	0	1
Gender	Female	Male
Hormonal Changes	Normal	Postmenopausal
Family History	No	Yes
Body Weight	Normal	Underweight
Calcium Intake	Adequate	Low
Vitamin D Intake	Insufficient	Sufficient
Physical Activity	Active	Sedentary
Smoking	No	Yes
Alcohol Consumption	Moderate	None
Medications Use	Corticosteroids	None
Prior Fractures	No	Yes

4. Data Descriptive Analysis

4.1 Target Variable (Osteoporosis)

The target variable that we want to study is the status of having osteoporosis, which is a binary variable. In the following there is its distribution chart.

- Yes: 61.9%
- No: 38.1%

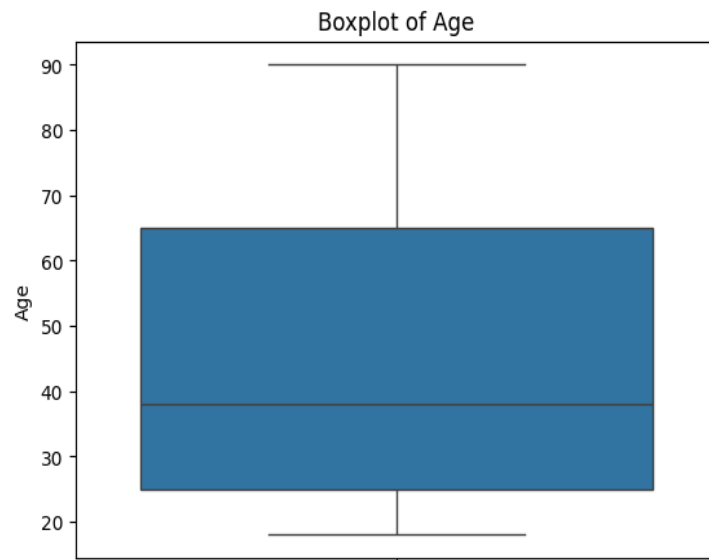
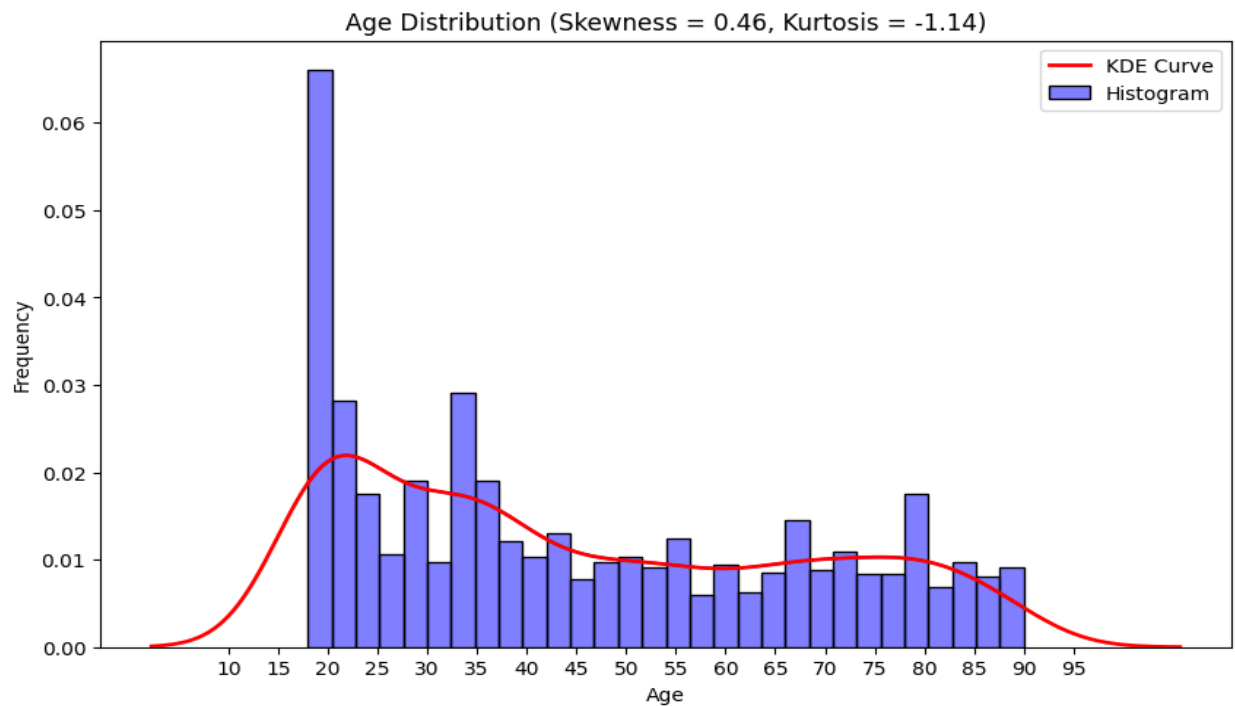


As it is obvious, the dataset is slightly imbalanced, with more participants classified as having osteoporosis. This distribution indicates a strong presence of the condition in the population studied. therefore, as well as calculating the distribution of other variables, we will study their conditional distribution under the status of osteoporosis.

4.2 Numeric Variable (Age)

The Age variable is a critical numerical feature that likely influences the risk of osteoporosis.

In the following, the distribution bar charts and the boxplot are shown.



From the above boxplot, we can notice some basic statistical parameters that we summarize it in the following table.

Min	Q1	Median	Q3	Max
18	25	38	65	90

The variable Age for the entire dataset shows a wide range of values from a minimum of 18 years to a maximum of 90 years ($\text{Range} = 90 - 18 = 72$). The first quartile (Q1) is 25 years, indicating that 25% of the participants are aged 25 or younger. The median age is 38 years, meaning that half of the participants are younger than 38. The third quartile (Q3) is 65 years, showing that 75% of the participants are aged 65 or younger. In addition, by studying quartile spread ($|Q_i - \text{Median}|$, $i=1,3$), the distance between the third quartile and the median is larger than the distance between the median and the first quartile. This asymmetry suggests a potential right skew.

Also, by calculating other statistical parameters we can see that the mean, mode and standard deviation of the age is respectively 45.04 and 22.31 and 34. The higher mean compared to the median and mode suggests a slight right skew toward older ages too, and the standard deviation of 22.31 reflects significant variability, indicating the dataset includes participants from a broad range of age groups.

We will now analyze the variable Age under the two conditions of having osteoporosis and not having it. This comparison will provide valuable insights into the relationship between age and osteoporosis prevalence.

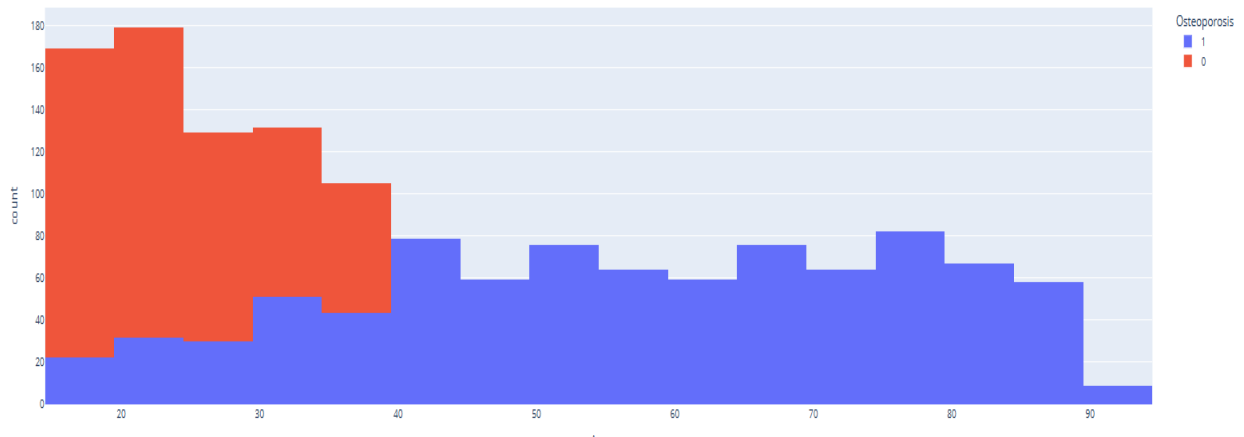
The mean is one of the most commonly studied statistical parameters in data analysis. Therefore, we begin our analysis by examining the mean.

[Mean]

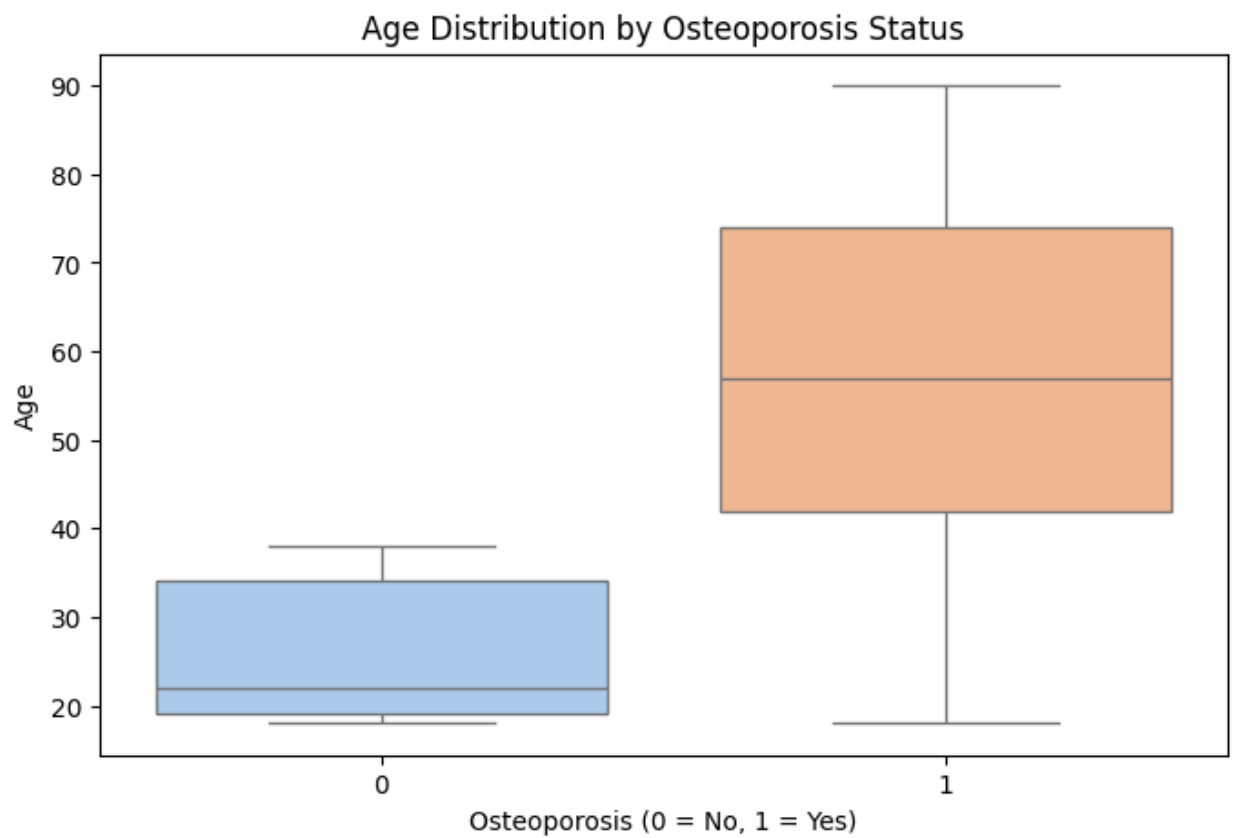
For participants without osteoporosis, the mean age is approximately 25 years, indicating that the majority of these individuals are relatively young.

For participants with osteoporosis, the mean age is approximately 57 years, showing that osteoporosis is more prevalent in older individuals.

The next graph shows the distribution of age based on presence and absence of osteoporosis with color.



Next, we will examine the boxplot charts for individuals with and without osteoporosis.



The table below summarizes the statistical characteristics of the Age variable, as illustrated in the boxplot.

osteoporosis	Min	Q1	Median	Q3	Max
0	18	19	22	34	38
1	18	42	57	74	90

[Range]

The minimum and maximum ages in both groups show that the age range for individuals without osteoporosis spans 20 years, while the range for those with osteoporosis is 72 years. This indicates a much broader age range for those with osteoporosis. Furthermore, the age range for people with osteoporosis fully encompasses the range of the first group, meaning there is no specific age interval where individuals are immune to osteoporosis.

[Median Age]

The median age for participants without osteoporosis is 22 years, meaning half of the participants in this group are younger than 22, and half are older.

The median age for participants with osteoporosis is 57 years, which aligns closely with the mean, indicating a relatively symmetric age distribution for this group.

[Interquartile Range (IQR)]

The interquartile range (IQR), calculated as $Q3 - Q1$, measures the range of the middle 50% of ages in each group.

For participants without osteoporosis, the IQR is 15 years ($34.0 - 19.0$), indicating a relatively narrow age distribution.

For participants with osteoporosis, the IQR is 32 years ($74.0 - 42.0$), reflecting a broader distribution of ages.

It is particularly noteworthy that the range between Q1 and Q3 for participants with osteoporosis is entirely higher than that of participants without osteoporosis. This clearly illustrates that the middle 50% of participants with osteoporosis are older compared to those without the condition.

In the table below, we also compare the standard deviation, mean, and mode in both states.

osteoporosis	STD	Mean	Mode
0	6.667574	25.234	34
1	19.679025	57.212	79

[Mode]

By comparing the modes, we observe that the most frequent age for participants without osteoporosis is **34 years**, while for those with osteoporosis, it is **79 years**. This stark difference highlights the key role of age in the development of osteoporosis, as older individuals are more frequently represented in the group with the condition.

[STD]

In the case of both age distributions (with and without osteoporosis), the standard deviation may not be the best measure of variability. This is because the distributions are not normal, as confirmed by both the **Shapiro-Wilk test** and **Normal test**:

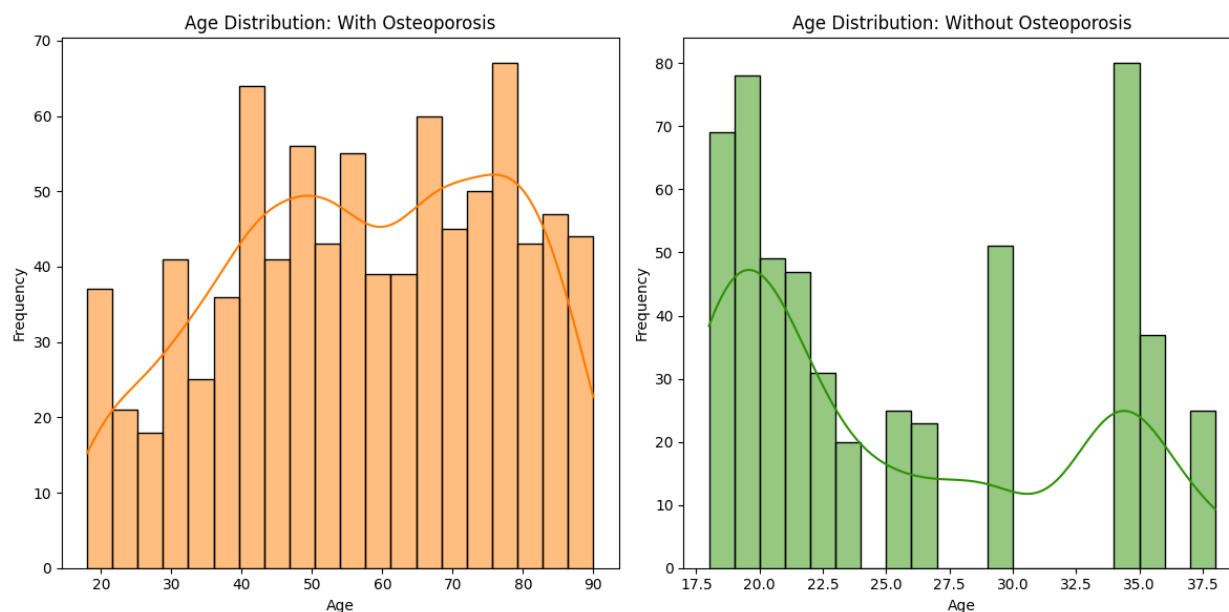
- For participants without osteoporosis, the p-value is 4.4×10^{-224} and 5.1×10^{-115} , respectively for Shapiro-Wilk and Normal test, which is far below the threshold of 0.05. This indicates the distribution significantly deviates from normality.
- Similarly, for participants with osteoporosis, the p-value is 8.4×10^{-148} and 7.5×10^{-45} , respectively, indicating significant non-normality.

Given these results, the standard deviation, which assumes symmetry and normality, is not a good measure for these distributions. Instead, measures like the interquartile range (IQR) are better suited for describing variability in these non-normal distributions, as they are less affected by skewness or outliers.

Significance Test for Age:

To assess the relationship between a numerical variable (Age) and a binary categorical variable (Osteoporosis), we can use either a t-test or a Mann-Whitney U test, depending on the distribution of the data. These statistical tests evaluate whether there is a significant difference between the distributions of two independent groups. The significance of the result is determined by the corresponding p-value, which indicates whether the observed difference is likely due to chance.

The independent t-test requires that the data in both groups be normally distributed. However, as observed from the standard deviation analysis and the graph below, the distribution of Age is not normal for either osteoporosis status.



Therefore, the Mann-Whitney U test is more appropriate in this case.

The results of the Mann-Whitney U test show a p-value of exactly 0, which is less than the standard significance threshold of 0.05. This indicates that the null hypothesis—that the distribution of Age is the same for both groups—can be rejected.

In conclusion, the test reveals a statistically significant difference in Age between individuals with and without osteoporosis. Specifically:

- The age distributions of the two groups are not identical.
- Older individuals are more likely to be associated with having osteoporosis.

This finding aligns with our previous analysis, which highlighted Age as a significant factor associated with the likelihood of developing osteoporosis.

In the following analysis, we will explore the relationships between other variables and osteoporosis to identify additional significant factors.

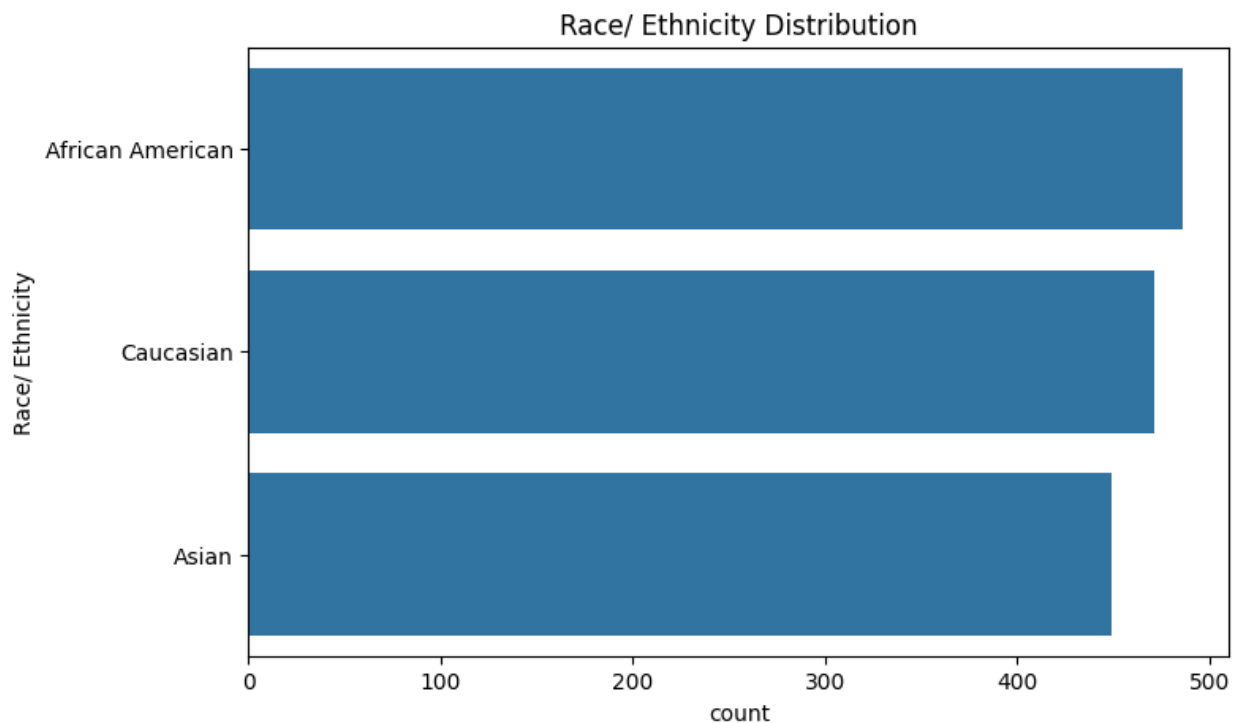
4.3 Categorical Variables

In this section, we will begin by examining the distribution of categorical variables. Then, we will assess each variable in relation to the presence or absence of osteoporosis and we will use the chi-squared test to determine whether there is a significant relationship between each variable and osteoporosis.

Race/Ethnicity Distribution:

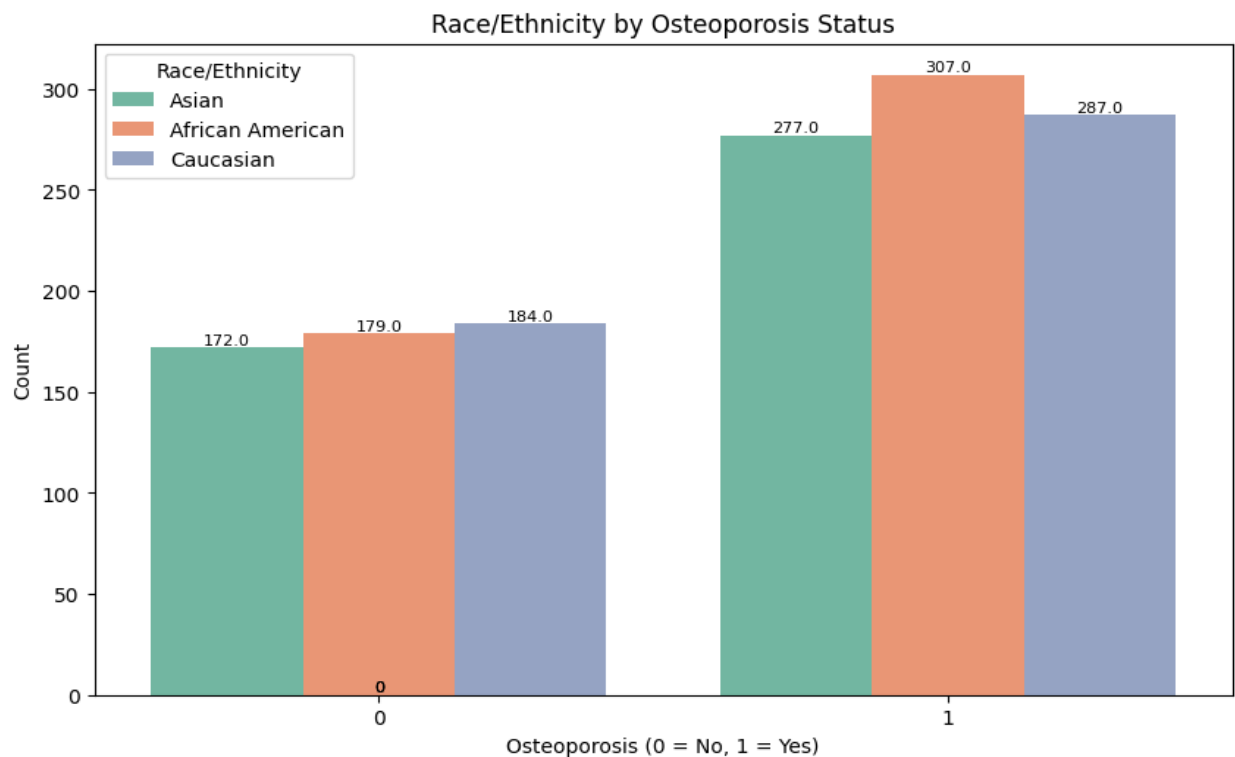
The dataset has a Very balanced distribution across ethnic groups, allowing for reliable comparisons between racial groups

- African American: 34.6%
- Caucasian: 33.5%)
- Asian: 31.9%



Race/Ethnicity Based on Osteoporosis Status:

Among individuals with osteoporosis, African Americans have the highest count (307), followed by Caucasians (287) and Asians (277). In the other hand, among those without osteoporosis, the number of Asians is lower, while Caucasians have the highest count. However, these differences are minor. The counts alone do not indicate a meaningful relationship between race and the likelihood of developing osteoporosis. Therefore, a statistical test, such as the chi-square test, should be performed to assess this association, leading to a more reliable conclusion.



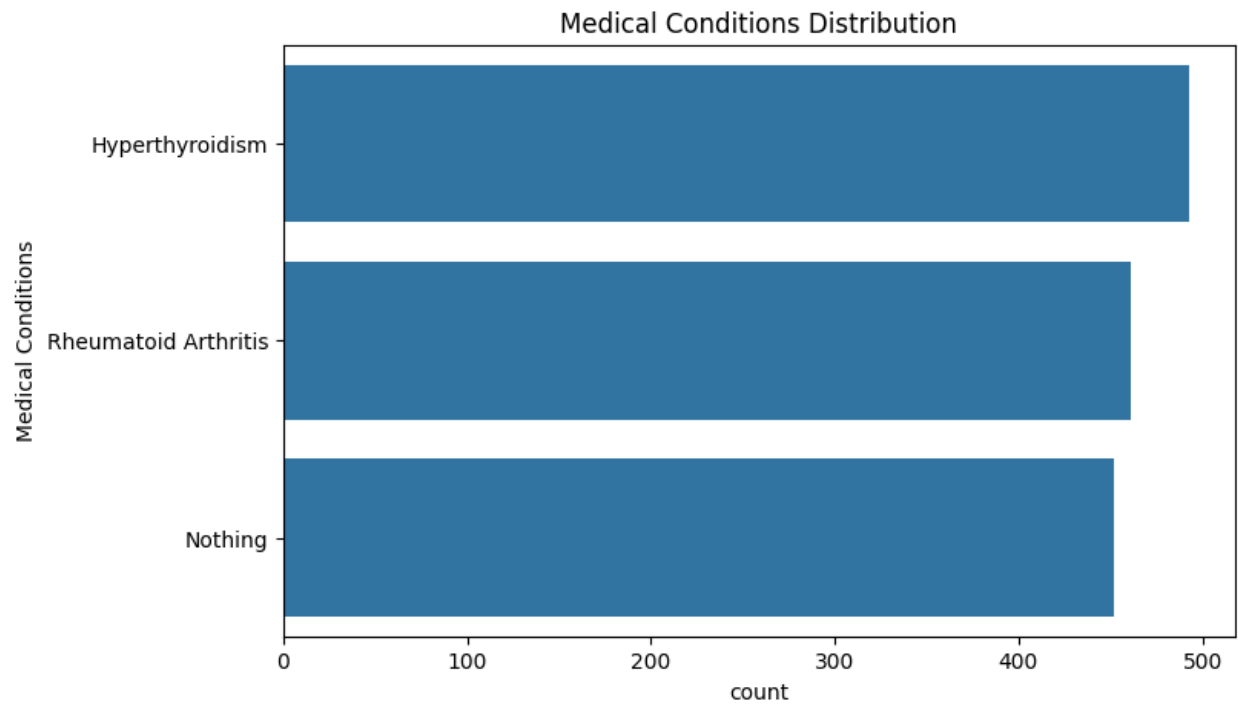
Statistical Significance:

The chi-square test result for Race/Ethnicity ($p=0.7691$) indicates no statistically significant association between race/ethnicity and osteoporosis in this dataset. This means the observed differences in counts are likely due to chance rather than a meaningful relationship.

Medical Conditions Distribution:

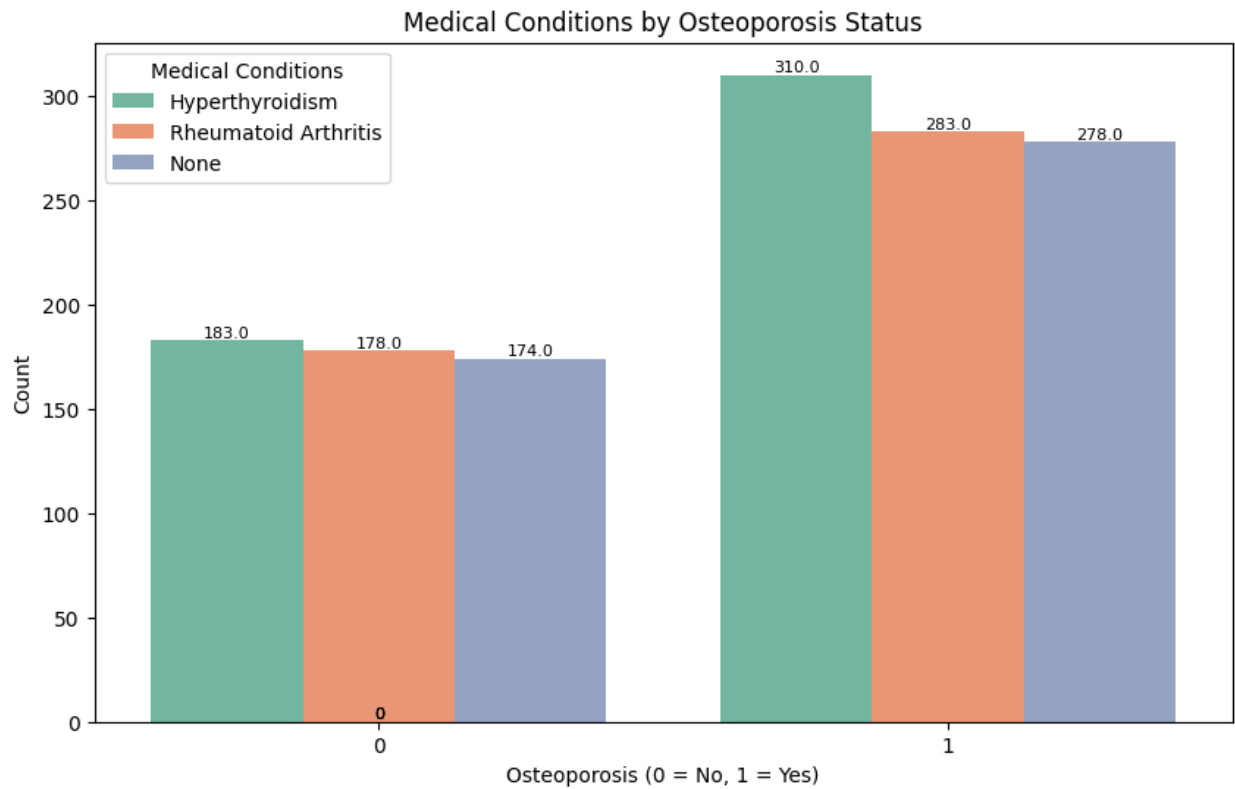
The distribution shows that a substantial proportion of participants have either hyperthyroidism or rheumatoid arthritis, while those categorized as 'Nothing' do not report any medical condition.

- Hyperthyroidism: 35.1%
- Rheumatoid Arthritis: 32.8%
- Nothing: 32.1%



Medical Conditions Based on Osteoporosis Status:

Among individuals with osteoporosis, those with hyperthyroidism (310) have higher counts compared to those with rheumatoid arthritis (283) or no medical conditions (278). Similarly, among individuals without osteoporosis, the distribution across the three categories is relatively even.



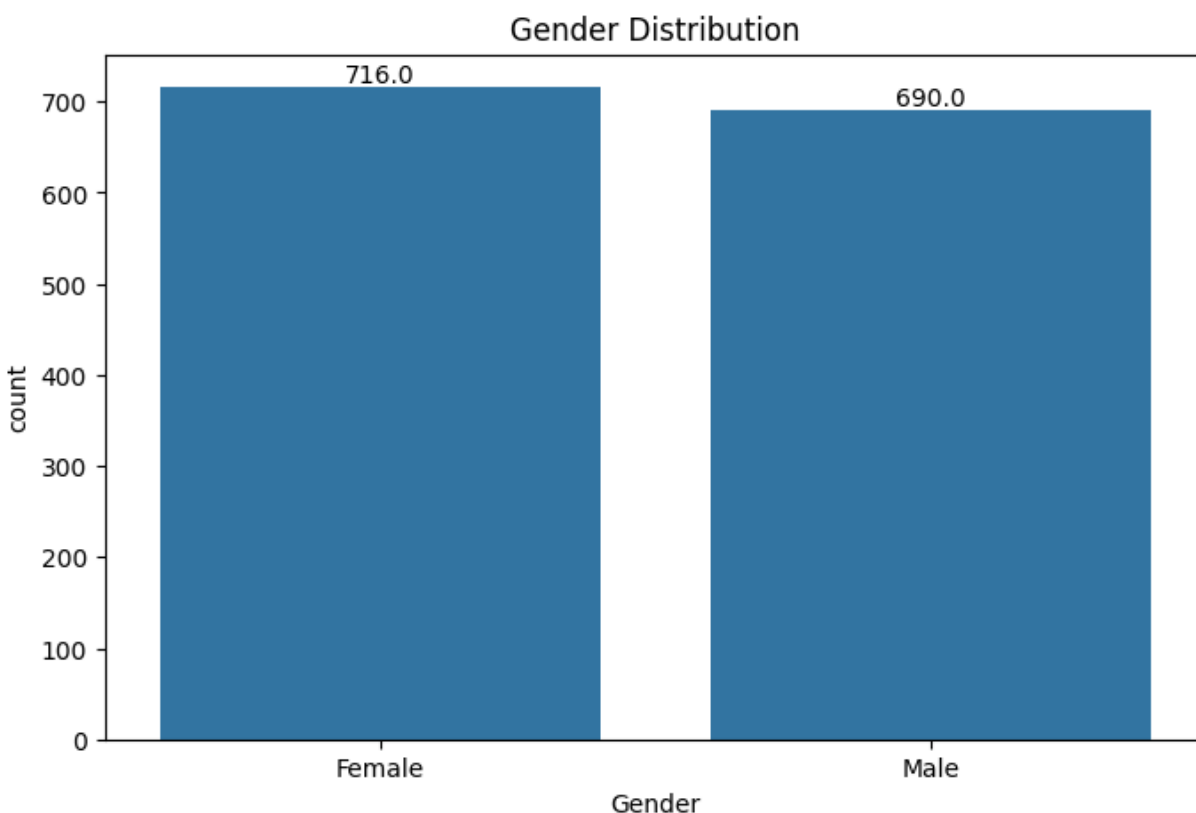
Statistical Significance:

The chi-square test shows a p-value of 0.8690. Therefore, despite higher counts of individuals with hyperthyroidism and rheumatoid arthritis among those with osteoporosis, the chi-square test confirms no significant association between medical conditions and osteoporosis.

Gender Distribution:

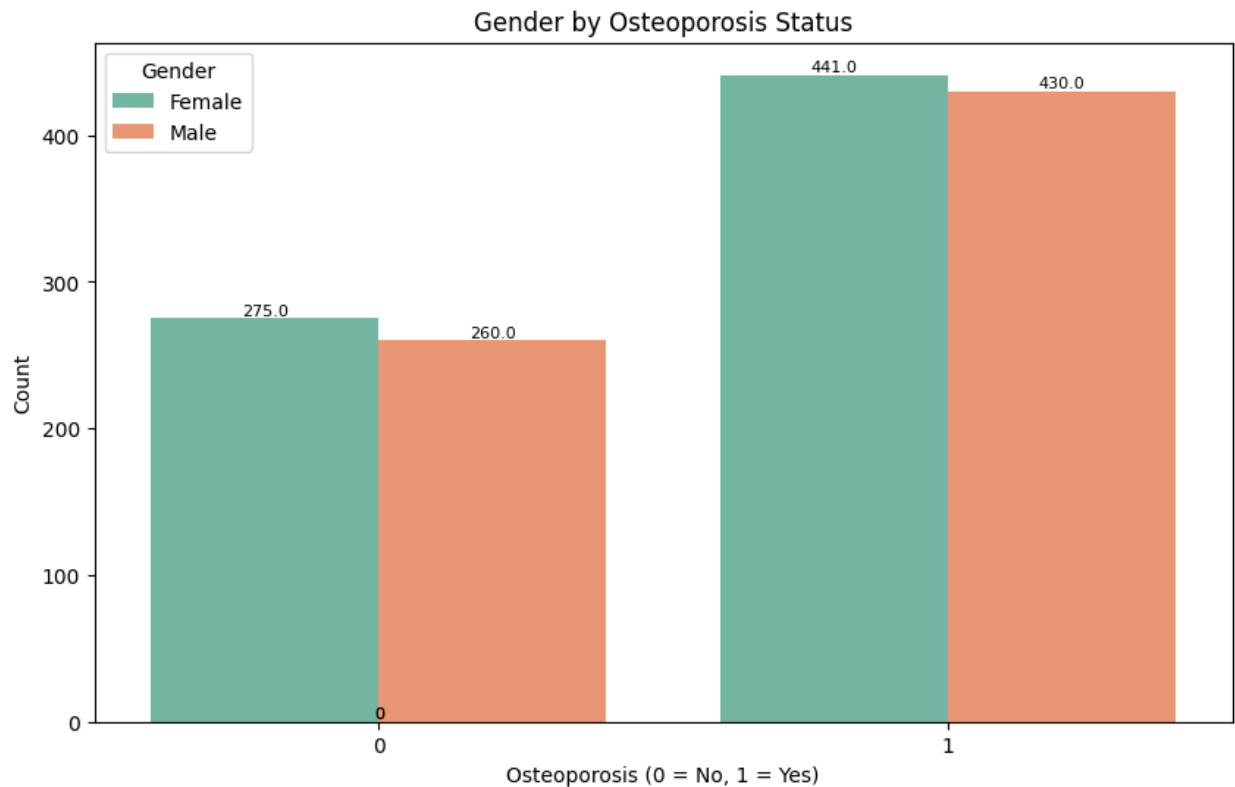
The dataset is relatively balanced in terms of gender, with a slightly higher proportion of females.

- Females: 50.9%
- Males: 49.1%



Gender Distribution Based on Osteoporosis Status:

When examining it based on the presence or absence of osteoporosis, we observe that the distribution of osteoporosis status is nearly identical for males and females, with slightly more females (441) than males (430) among those with osteoporosis. Similarly, among those without osteoporosis, the counts are close for females (275) and males (260).



Statistical Significance:

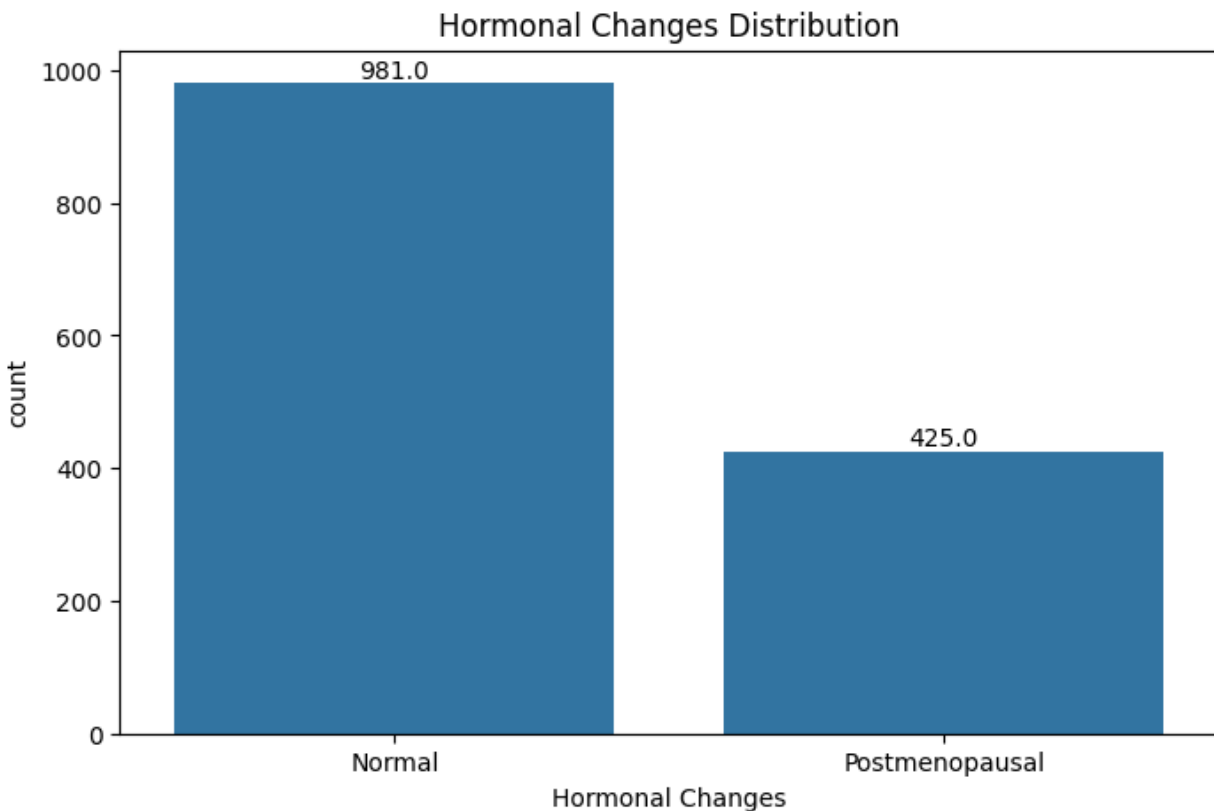
The chi-square test ($p=0.8215$) indicates no significant association between gender and osteoporosis. This suggests that the observed differences in counts are likely due to chance.

Therefore, as the bar chart shows similar distributions of osteoporosis status for males and females in both statuses, the chi-square test also confirms that gender is not significantly associated with osteoporosis in this dataset.

Hormonal Changes Distribution:

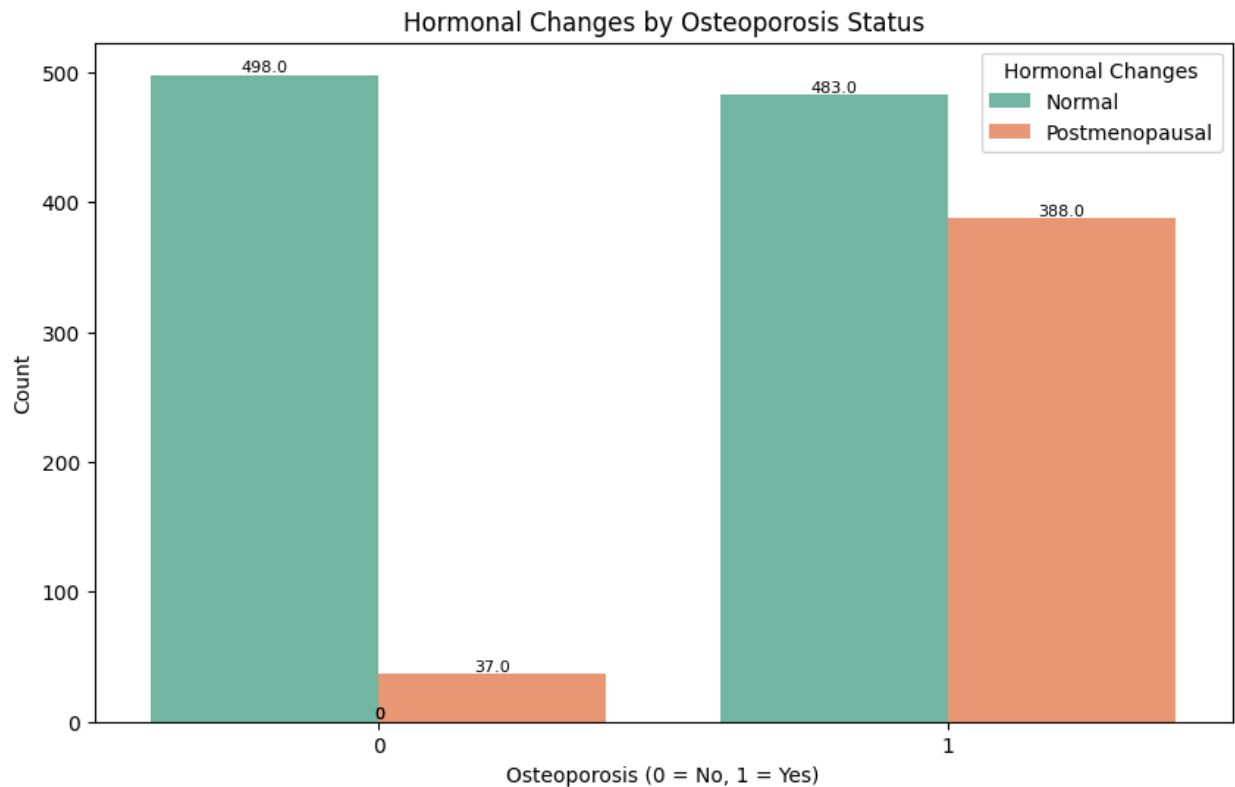
A majority of participants do not report hormonal changes, but about 30% do. Therefore, it shows significant portion of postmenopausal cases.

- Normal: 69.8%
- Postmenopausal: 30.2%



Hormonal Changes Based on Osteoporosis Status:

The distribution of hormonal changes shows a striking difference between individuals with and without osteoporosis. Among those without osteoporosis, the majority (498) are categorized as having normal hormonal levels, while a significantly smaller proportion (37) are postmenopausal. Conversely, for individuals with osteoporosis, the distribution shifts notably. 483 have normal hormonal levels, but the count for postmenopausal individuals rises sharply to 388.



Statistical Significance:

This substantial increase in the postmenopausal category among those with osteoporosis highlights a potential link between postmenopausal status and the prevalence of osteoporosis. This observation is further supported by the results of the Chi-square test, which yielded a statistic of 220.75 and a p-value of 0.0000. These findings indicate a statistically significant association between hormonal changes and osteoporosis.

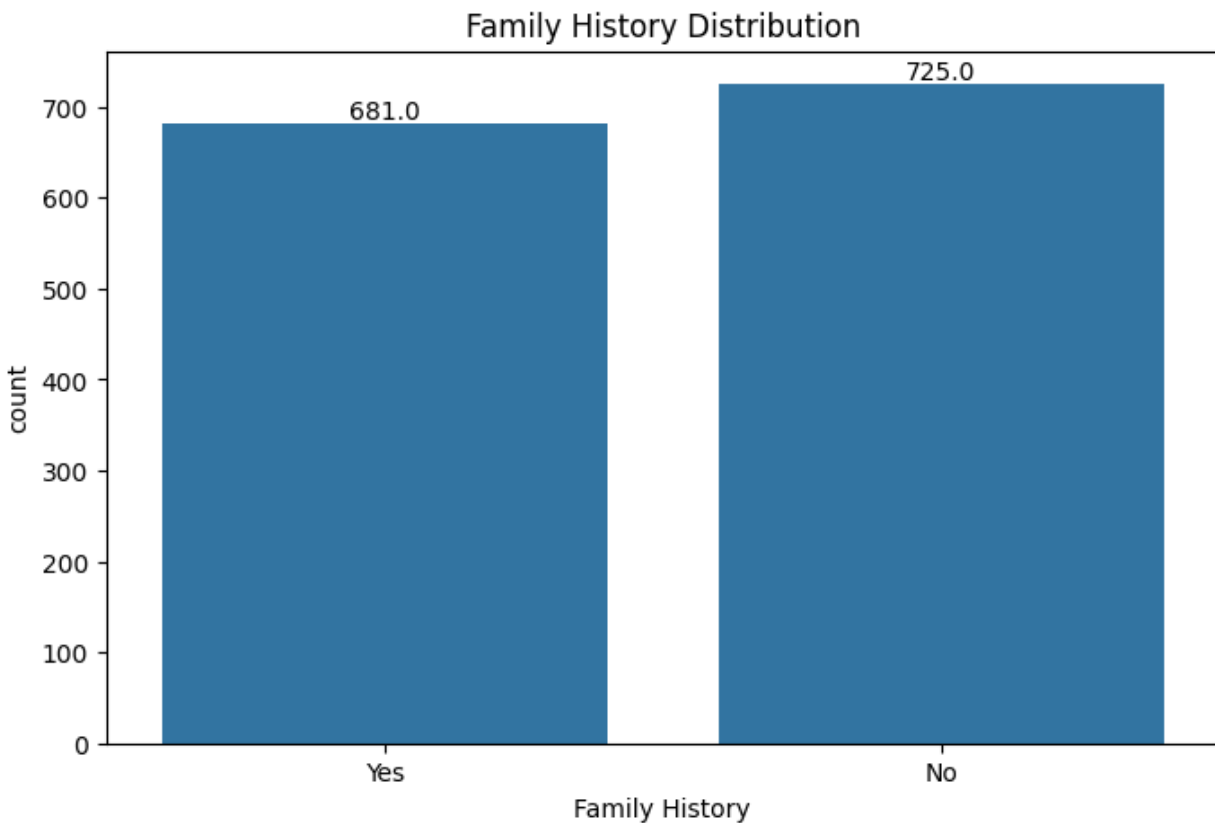
In summary, while normal hormonal levels dominate in both groups, the prevalence of postmenopausal status is markedly higher among those with osteoporosis.

This suggests that hormonal changes, particularly postmenopausal status, play a crucial role in the development of osteoporosis.

Family History Distribution:

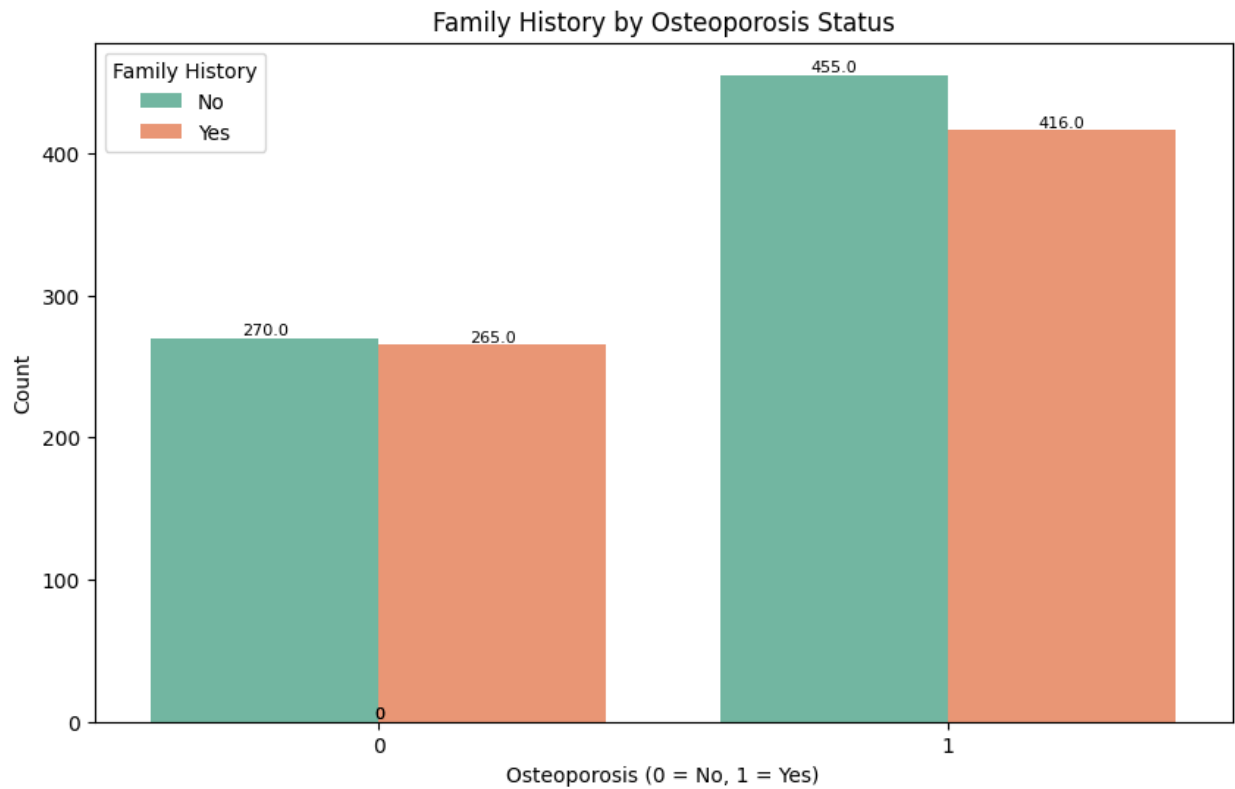
The dataset shows a near-even split between individuals with a family history of osteoporosis and those without.

- No family history: 51.6%
- Yes family history: 48.4%



Family History Based on Osteoporosis Status:

The distribution of "Yes" and "No" family history is similar across osteoporosis statuses. There is no noticeable difference in the heights of the bars for osteoporosis equals 0 and 1.



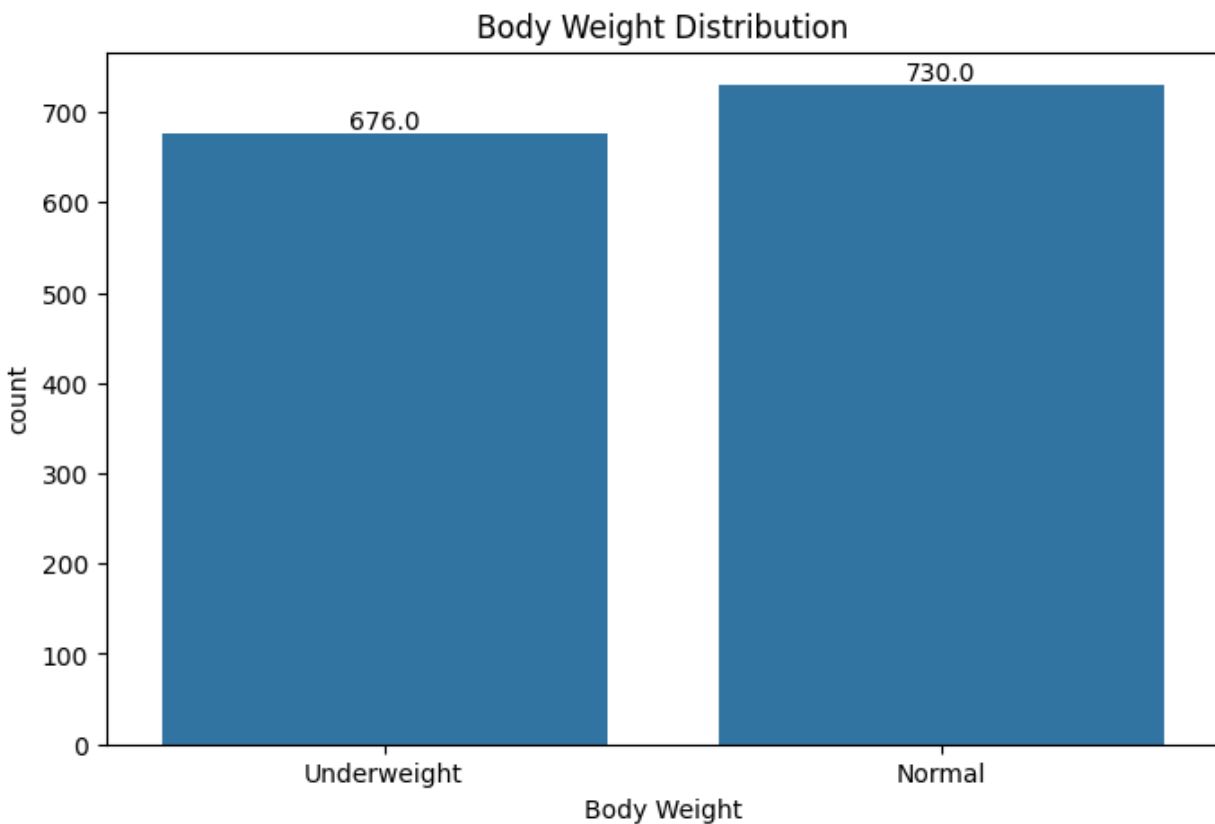
Statistical Significance:

The chi-square test ($p=0.5549$) indicates no significant association between family history and osteoporosis

Body Weight Distribution:

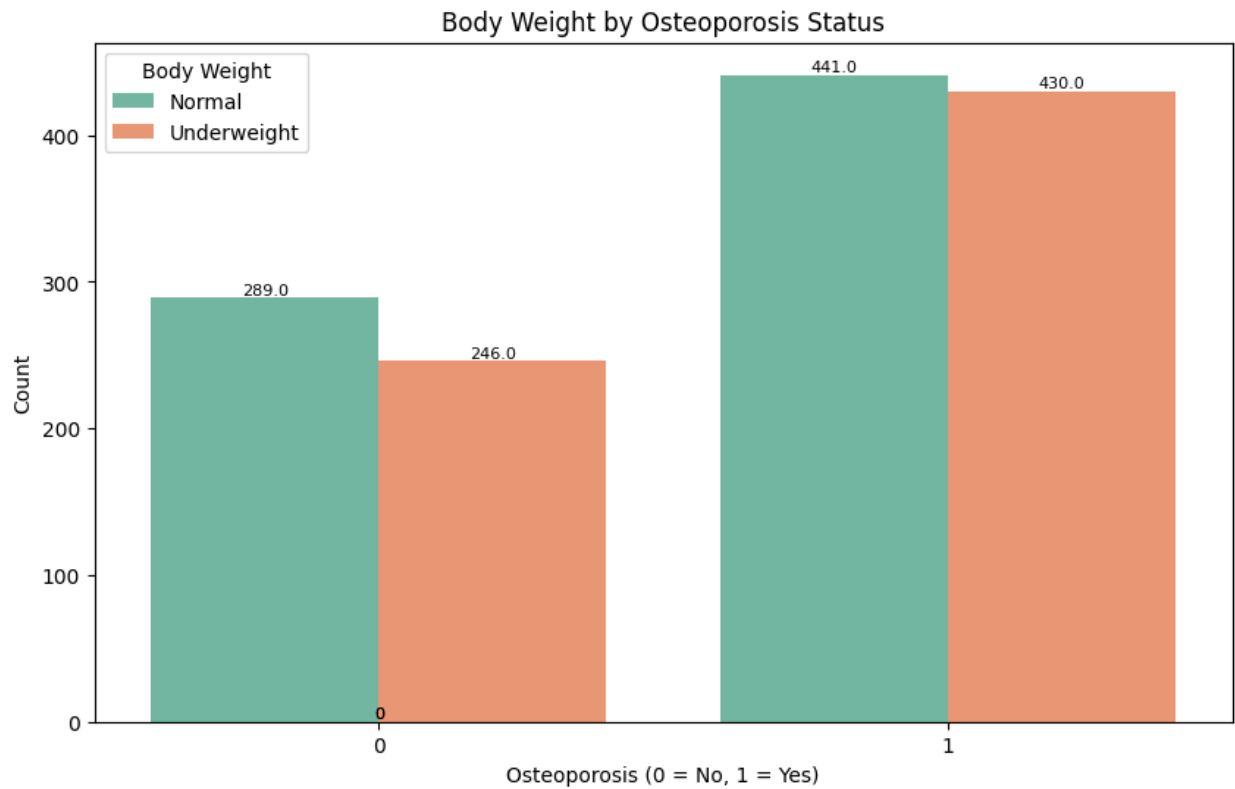
Body weight is nearly evenly distributed, suggesting diverse body composition among participants.

- Normal weight: 51.9%
- Underweight: 48.1%



Body Weight Based on Osteoporosis Status:

The distribution of osteoporosis status across the body weight categories is similar. Among those with osteoporosis, the counts for "Normal" (441) and "Underweight" (430) are almost equal.



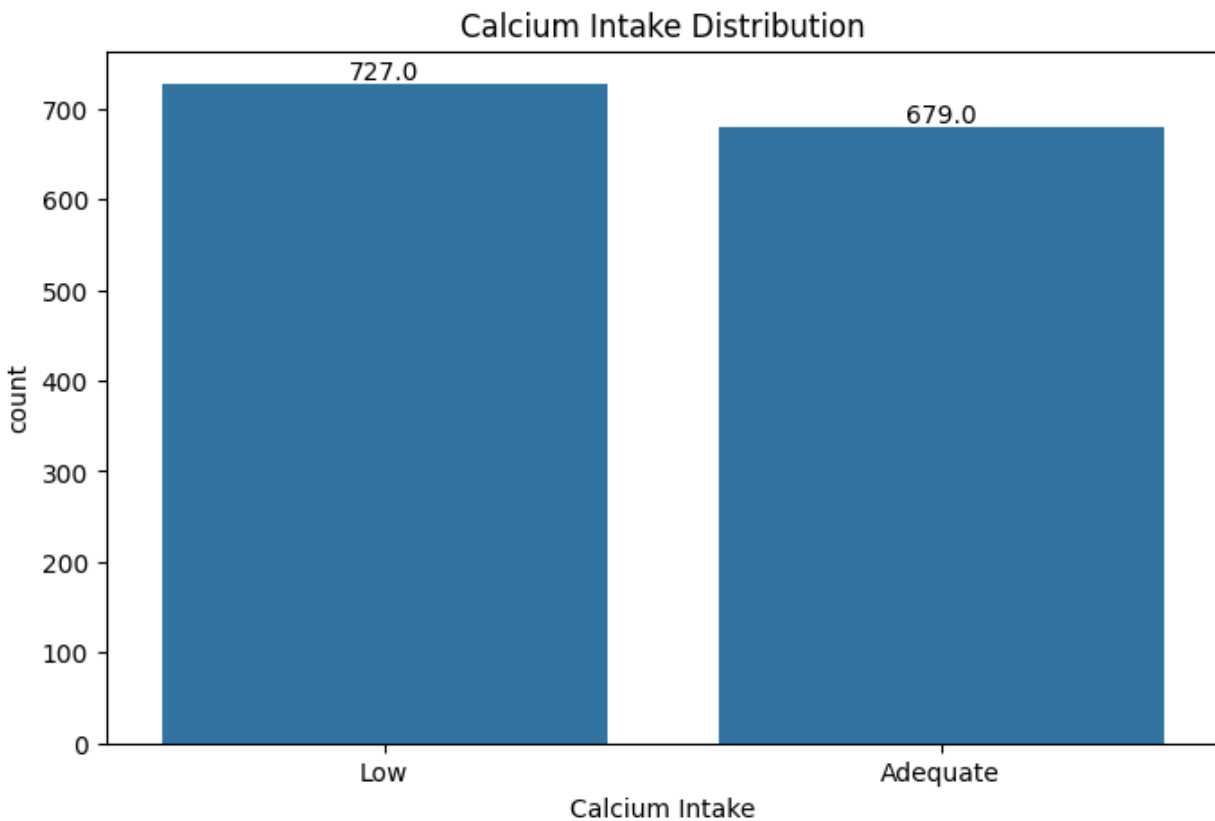
Statistical Significance:

In the result of chi-square test the p-value is equal to 0.2383 so both the bar chart and chi-square test suggest that body weight categories are not significantly associated with osteoporosis in this dataset.

Calcium Intake Distribution:

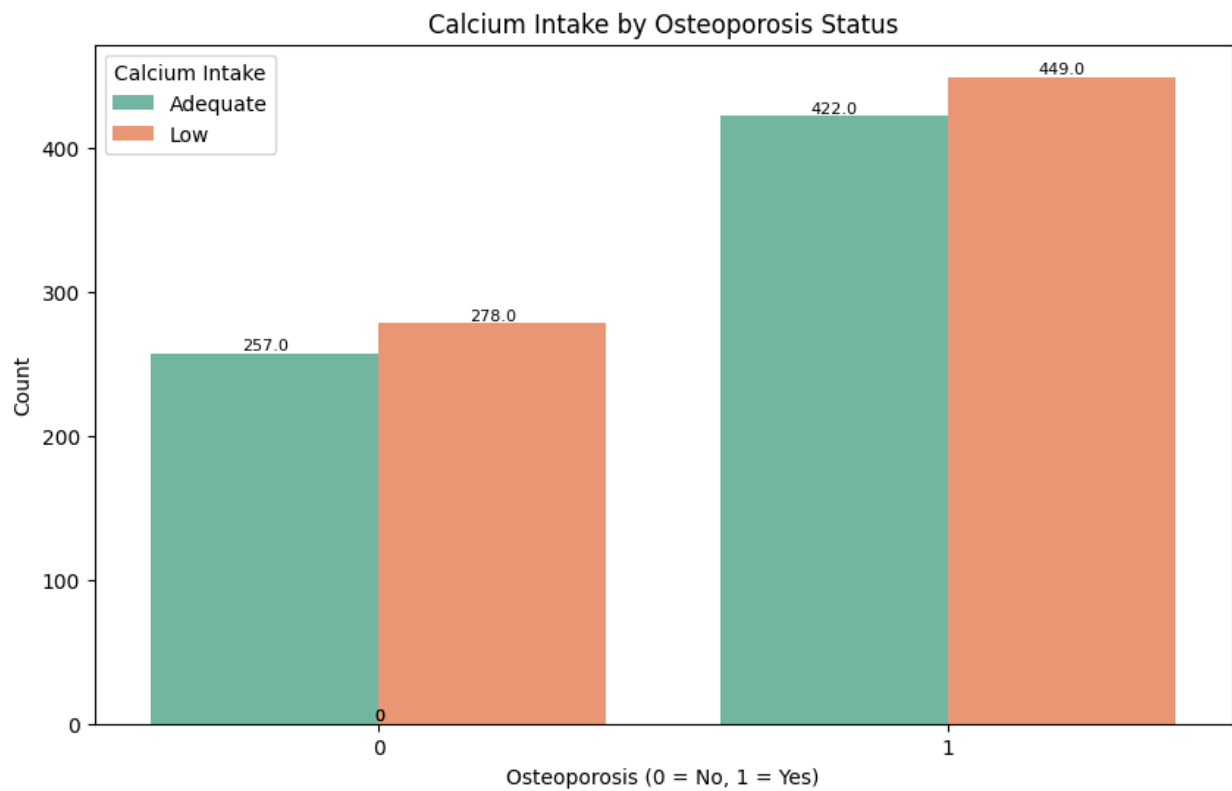
Good balance between intake levels

- Low intake: 51.7%
- Adequate intake: 48.3%



Calcium Intake Based on Osteoporosis Status:

Both "Adequate" and "Low" calcium intake categories are similarly represented across osteoporosis statuses. There is no clear visual difference in the bar heights.



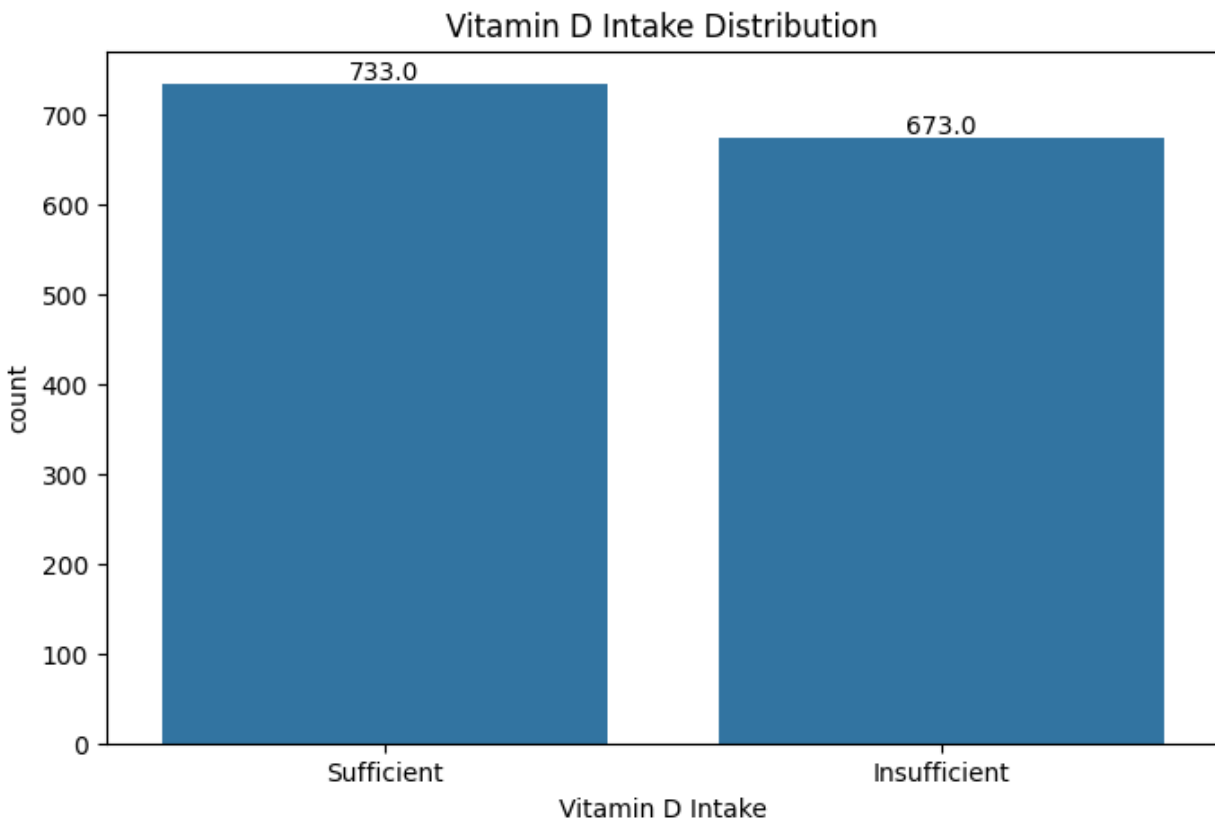
Statistical Significance:

The chi-square test ($p=0.9240$) indicates also no significant association between calcium intake and osteoporosis.

Vitamin D Intake Distribution:

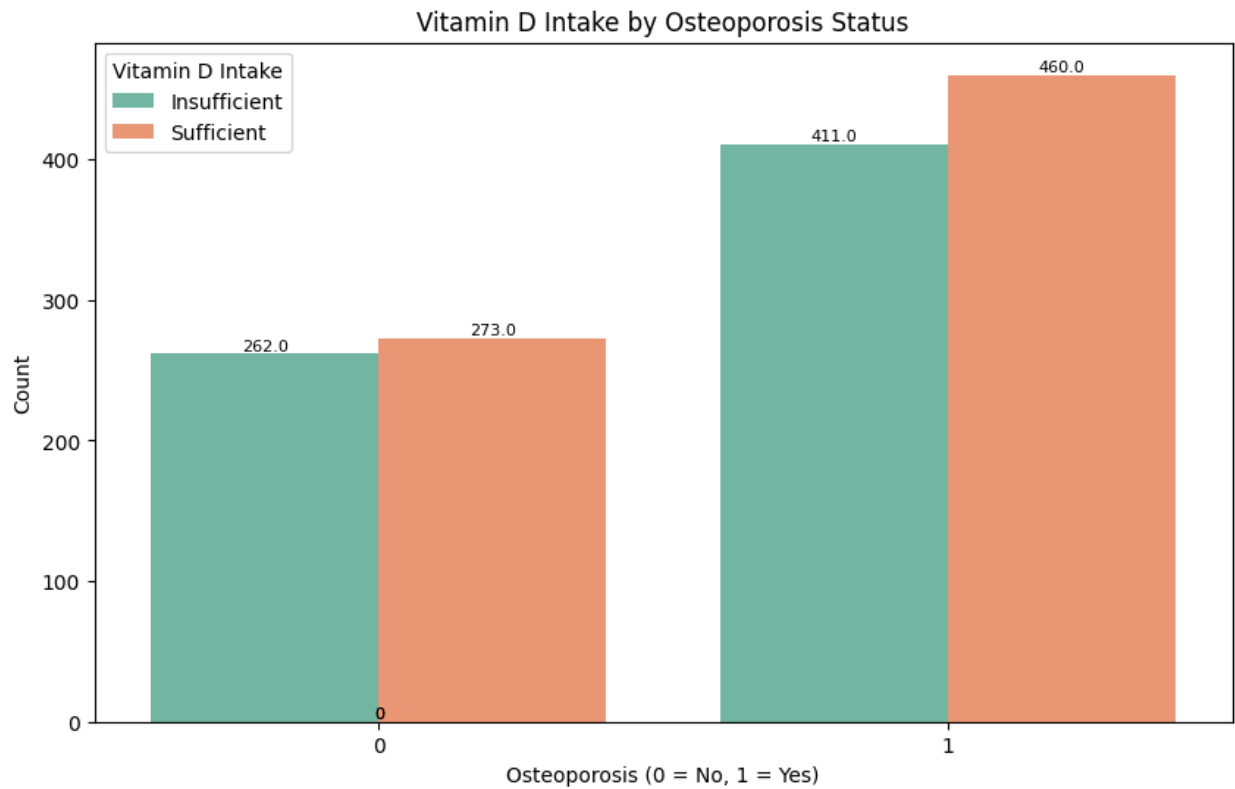
Similar to calcium intake, vitamin D levels are evenly distributed.

- Sufficient: 52.1%
- Insufficient: 47.9%



Vitamin D Intake Based on Osteoporosis Status:

The split between "Sufficient" and "Insufficient" vitamin D intake is nearly identical for osteoporosis = 0 and 1.



Statistical Significance:

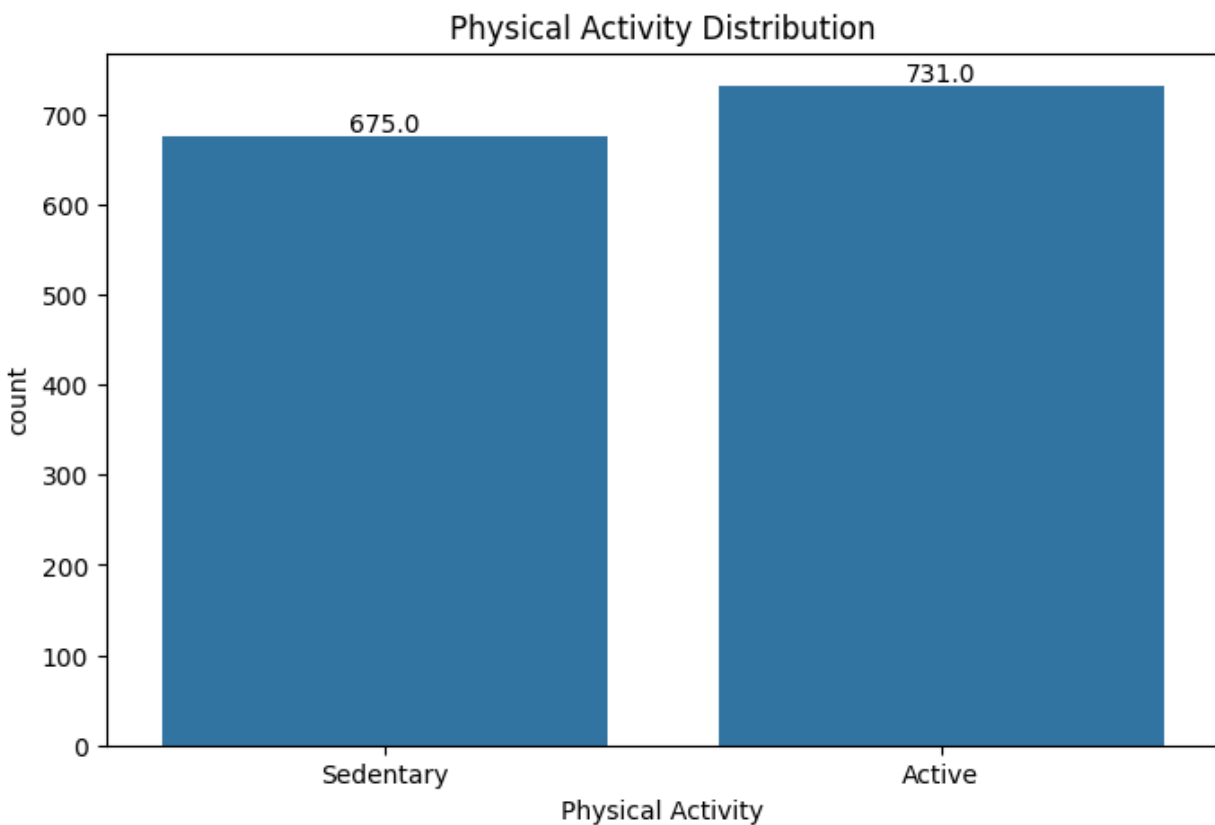
Chi-square Test: $p=0.5515$

So we can conclude that Vitamin D intake does not show a statistically significant association with osteoporosis.

Physical Activity Distribution:

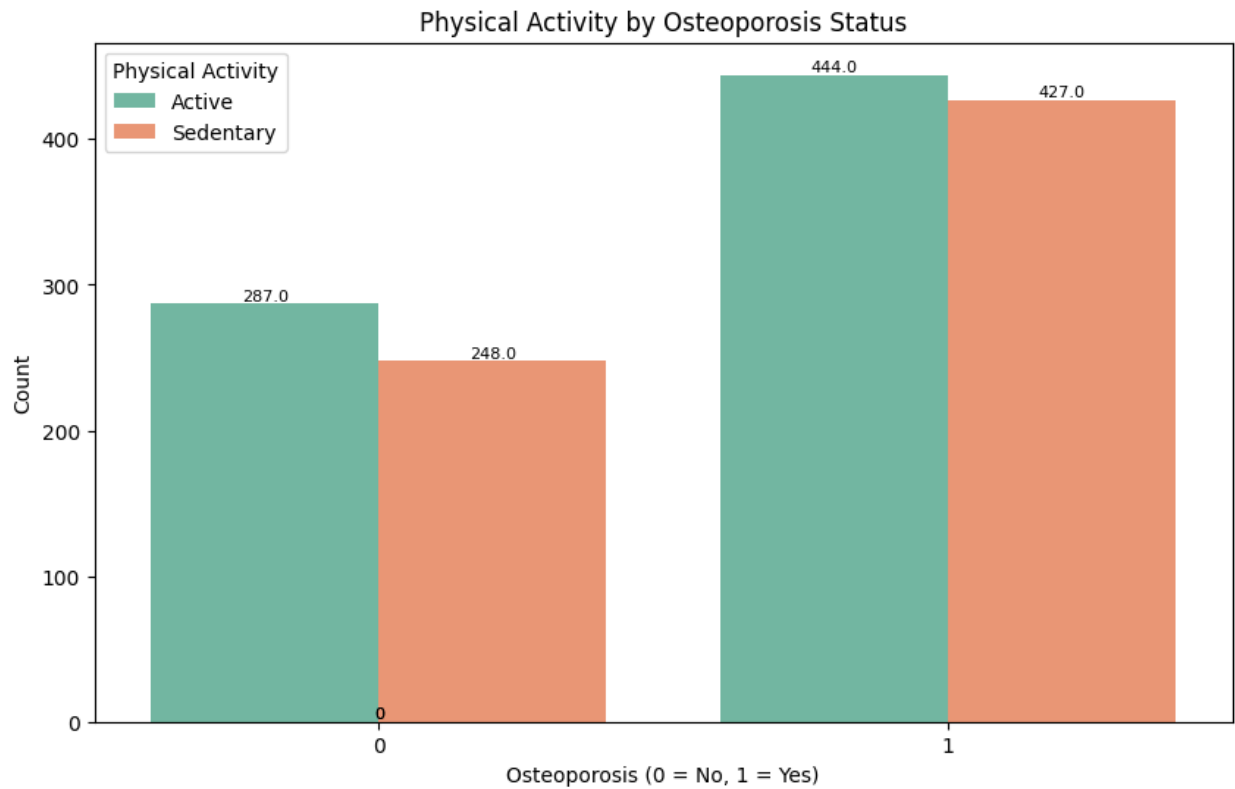
A balanced split in physical activity levels

- Active: 52%
- Sedentary: 48%



Physical Activity Based on Osteoporosis Status:

The proportion of "Active" and "Sedentary" individuals is nearly identical across osteoporosis statuses, with no apparent visual difference.



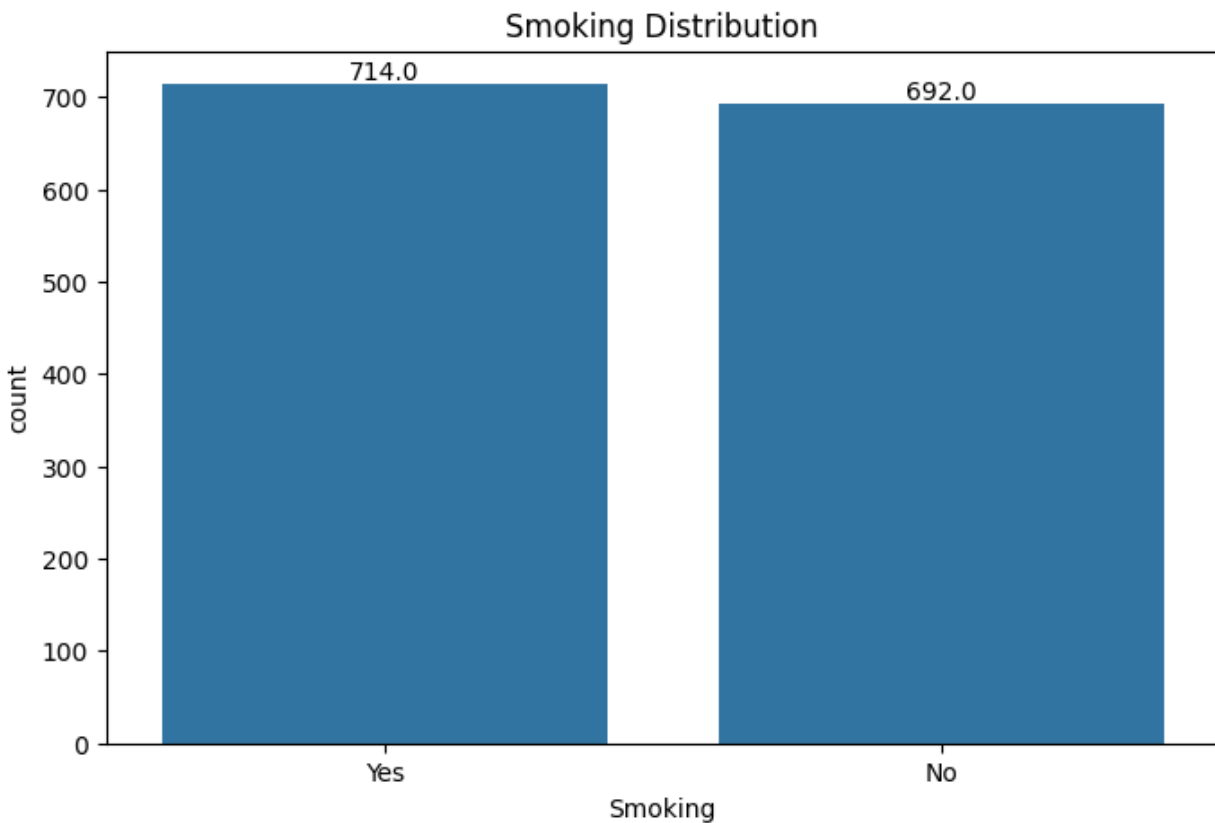
Statistical Significance:

The chi-square test ($p=0.3588$) indicates no significant association between physical activity and osteoporosis.

Smoking Distribution:

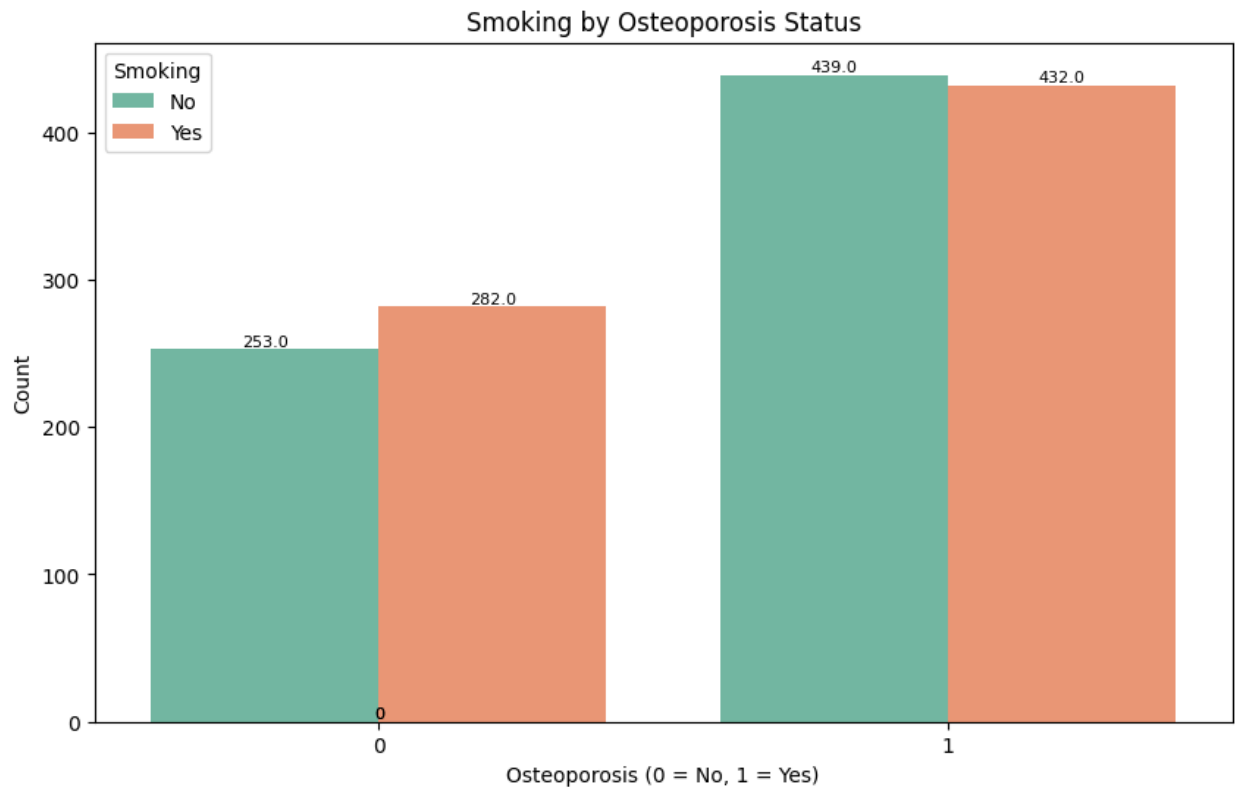
Smoking status is fairly balanced.

- Smokers: 50.8%
- Non-smokers: 49.2%



Smoking Based on Osteoporosis Status

Both "Yes" and "No" smoking groups are similarly distributed across osteoporosis statuses. The bar heights are comparable for each group.



Statistical Significance:

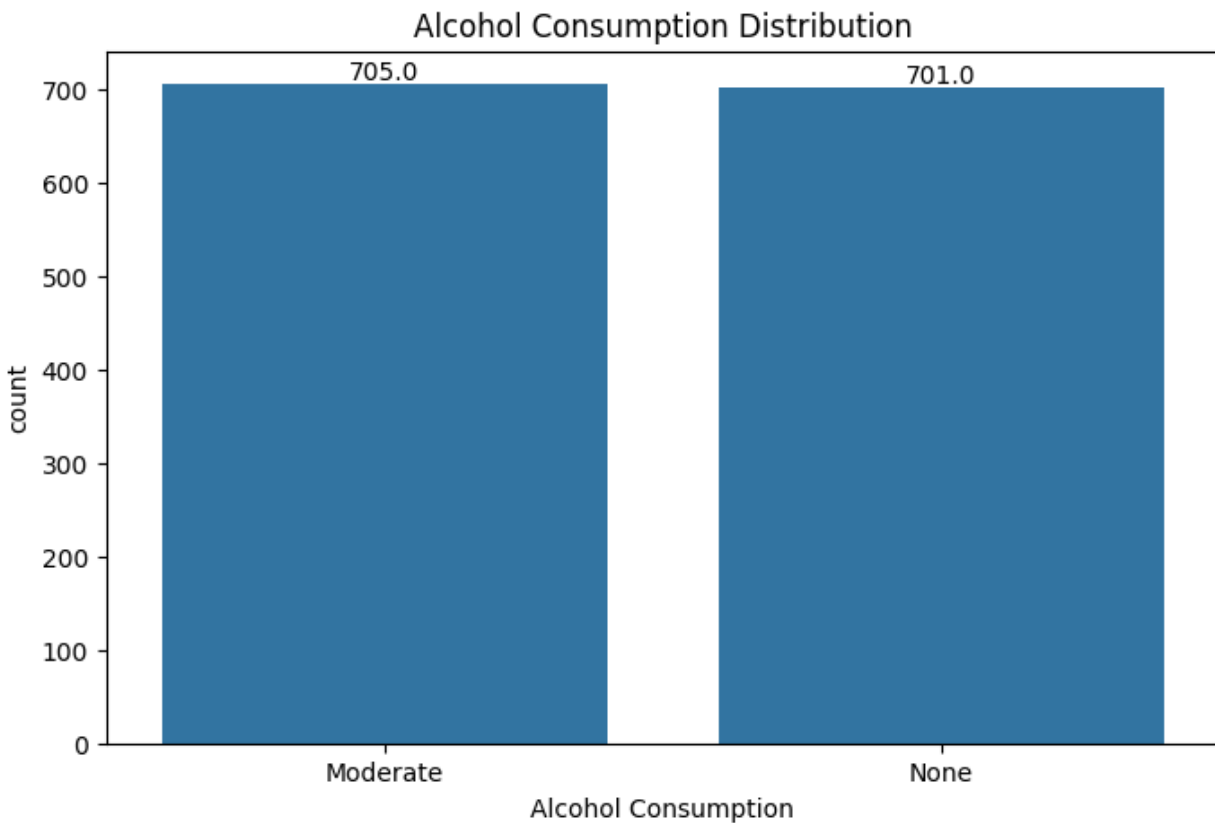
Chi-square Test: $p=0.2809$

Conclusion: Smoking does not show a statistically significant association with osteoporosis.

Alcohol Consumption Distribution:

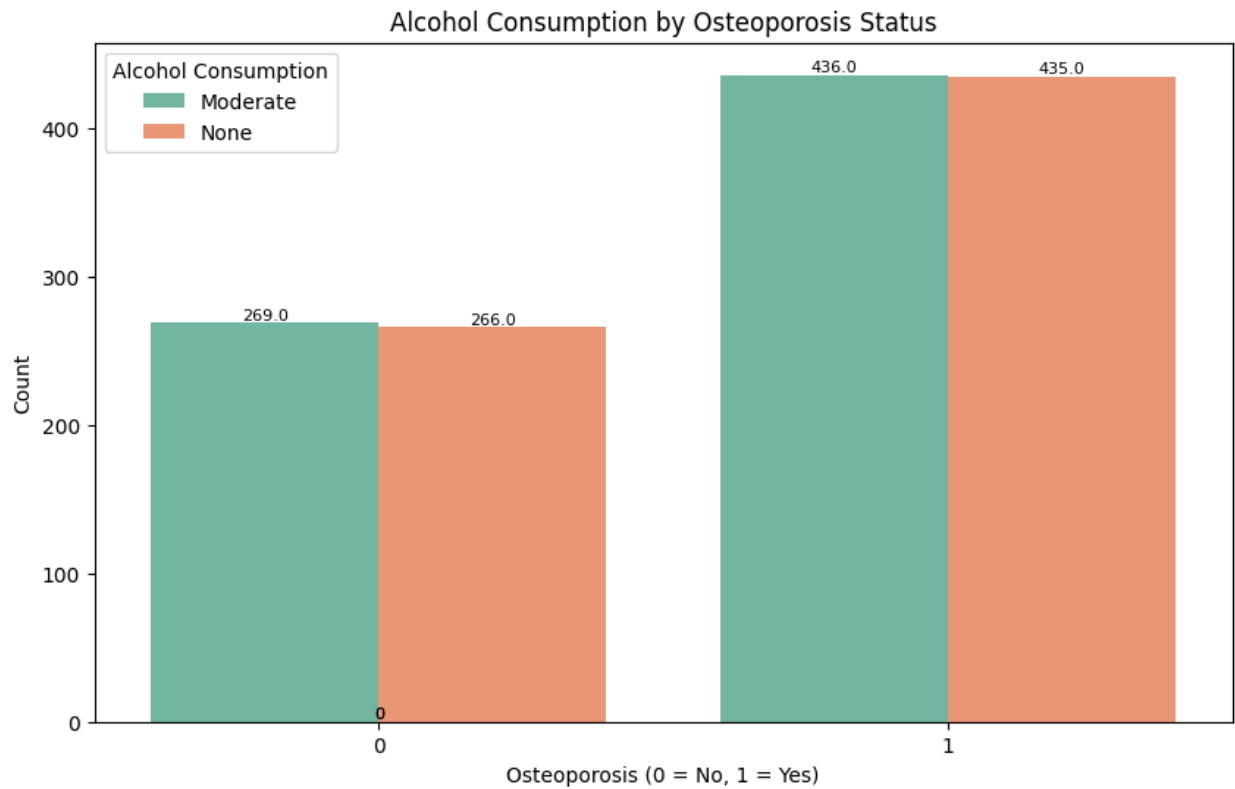
Alcohol consumption is evenly distributed.

- Moderate drinkers: 50.1%
- Non-drinkers: 49.9%



Alcohol Consumption Based on Osteoporosis Status:

Participants with "Moderate" and "None" alcohol consumption are equally represented across osteoporosis statuses.



Statistical Significance:

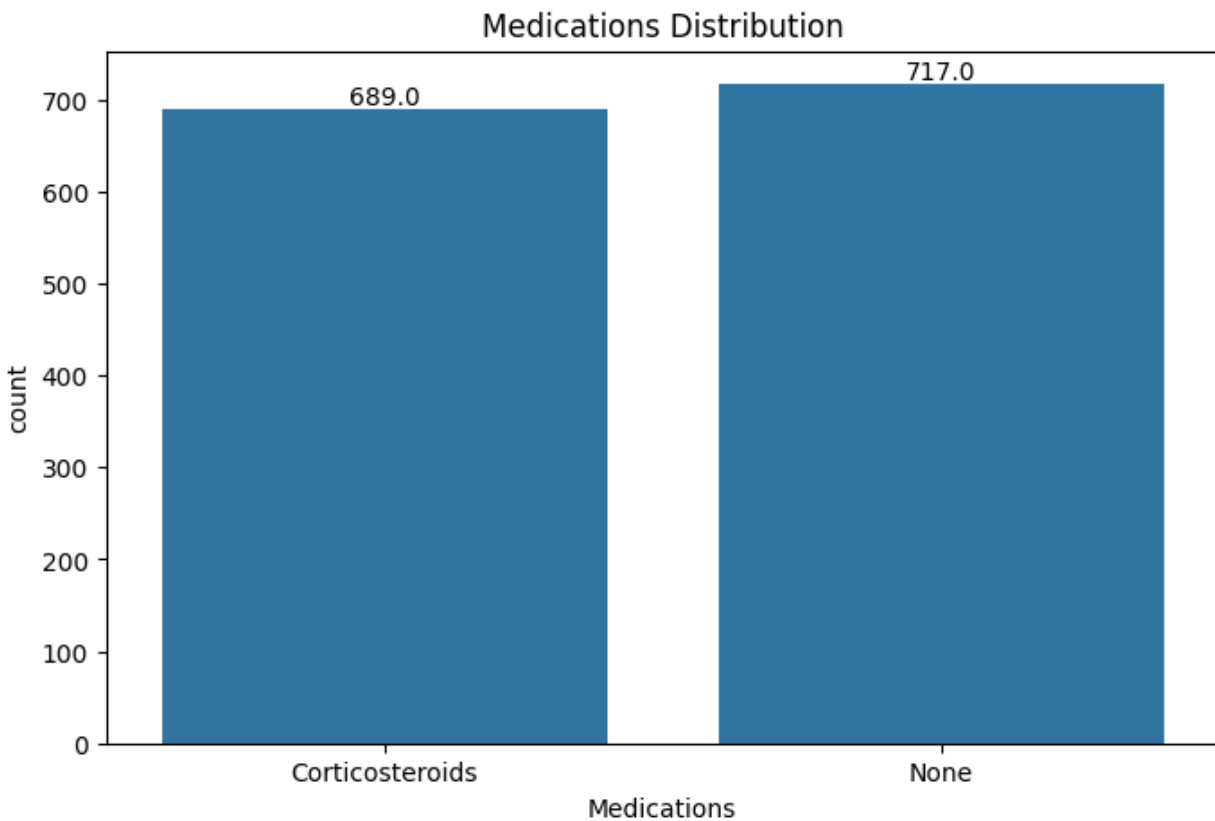
Chi-square Test: $p=0.9791$

Conclusion: Alcohol consumption does not have a statistically significant association with osteoporosis.

Medications Distribution:

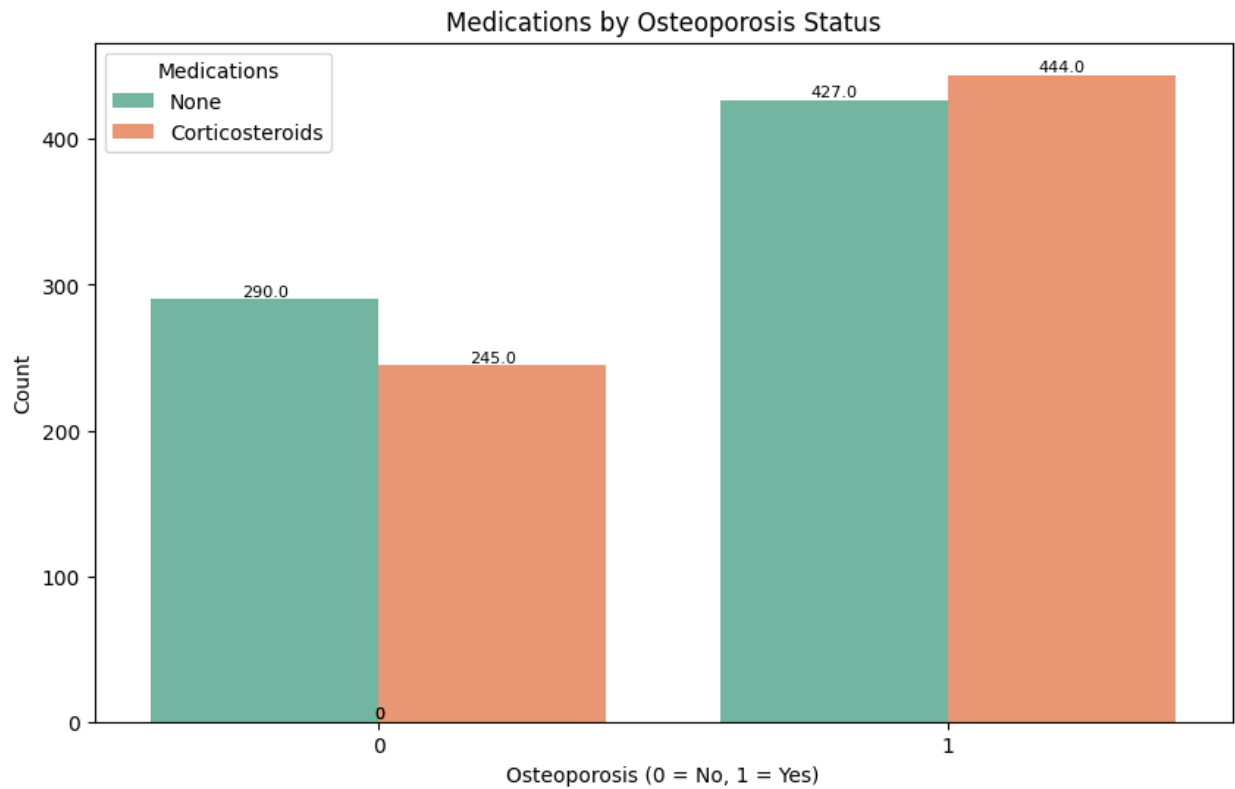
A near-even split indicates diverse medical interventions among participants.

- No medications: 51%
- Taking corticosteroids: 49%



Medications Based on Osteoporosis Status:

Participants using "Corticosteroids" are slightly more prevalent among osteoporosis = 1, but the difference is marginal. The bar heights are close in both statuses.



Statistical Significance:

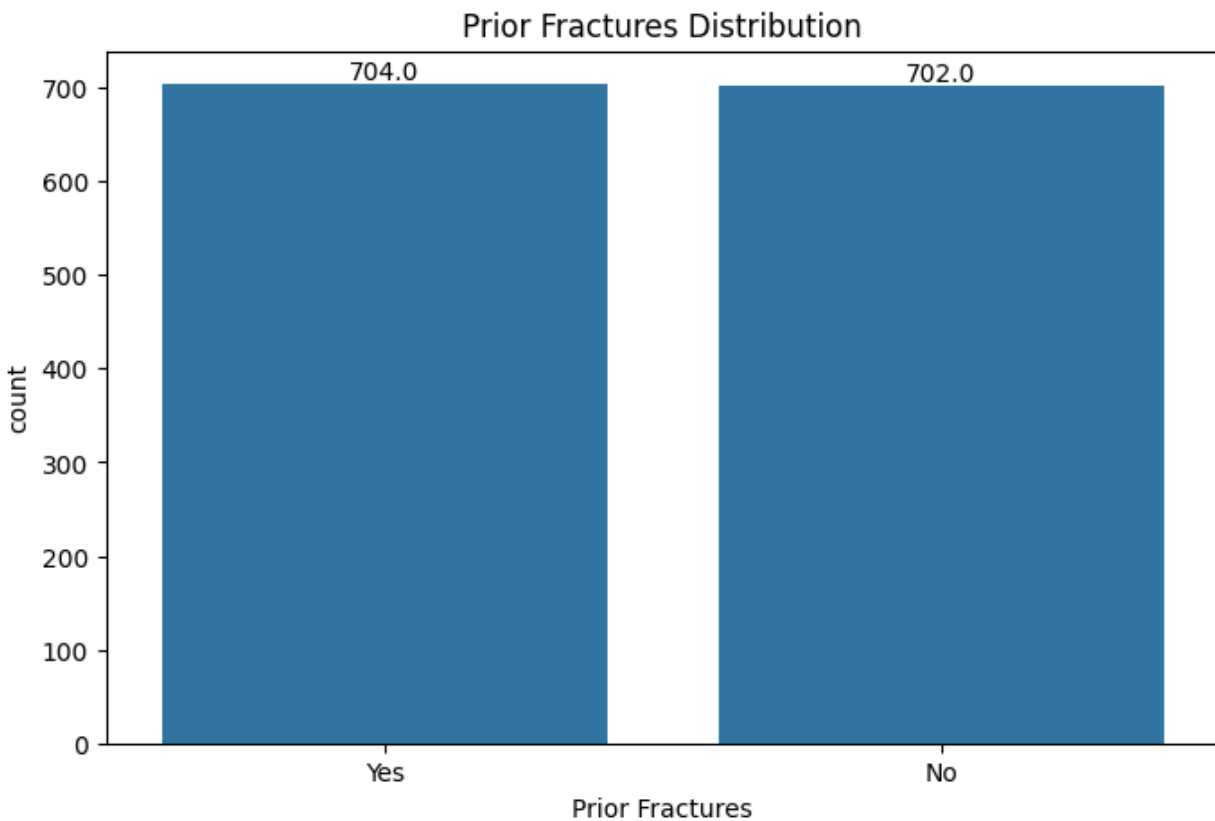
Chi-square Test: $p=0.0669$

Conclusion: Although there appears to be a trend, the chi-square test does not confirm a statistically significant association between medication use and osteoporosis

Prior Fractures Distribution:

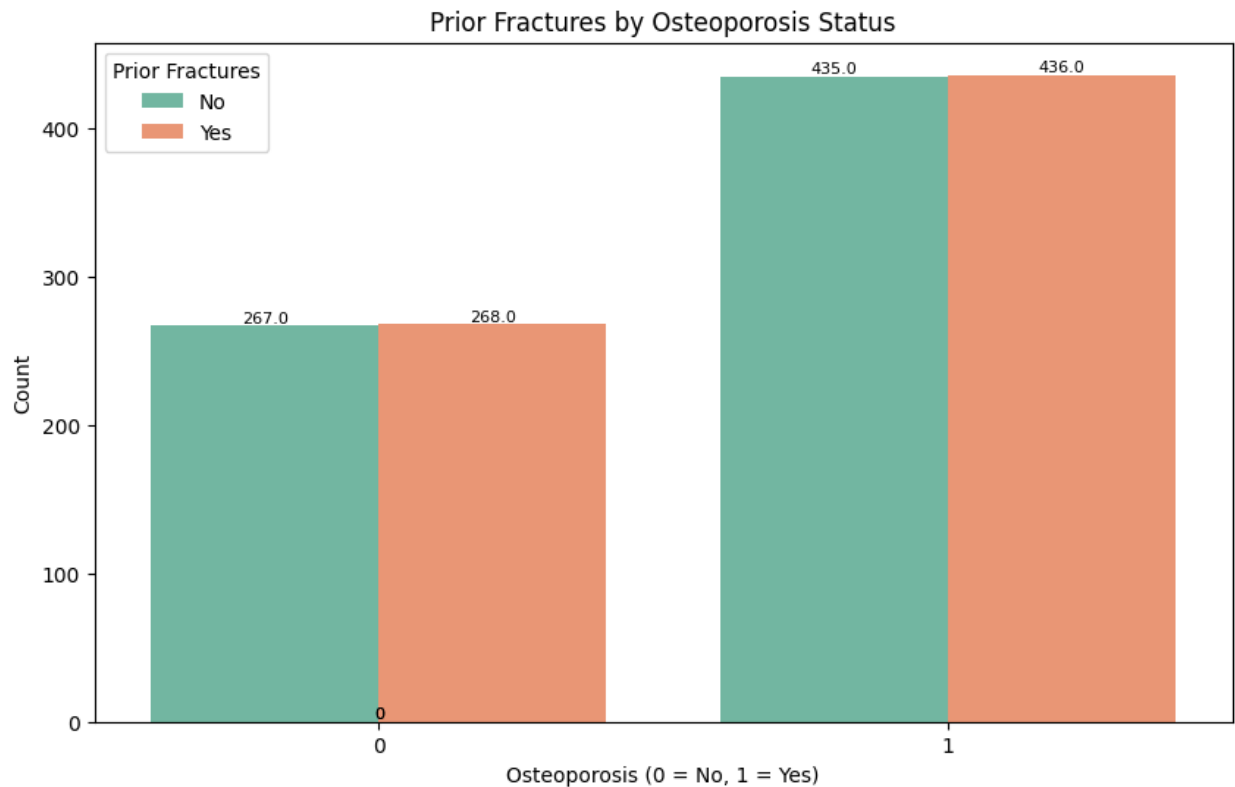
An almost equal distribution

- With prior fractures: 50.1%
- Without prior fractures: 49.9%



Prior Fractures Based on Osteoporosis Status:

Participants with prior fractures are equally represented across both osteoporosis statuses, with 436 individuals with fractures and 435 without fractures among those with osteoporosis.



Statistical Significance:

The chi-square test results show a p-value of exactly 1, which indicates that there is no statistically significant association between prior fractures and osteoporosis. This means that the likelihood of having osteoporosis does not differ based on the presence or absence of prior fractures in this dataset, so both the bar chart and the chi-square test results consistently suggest that prior fractures do not have a meaningful impact on osteoporosis status in this case.

The "Relationship Score" is a measure of effect size, indicating the strength of the association between each variable and target variable (osteoporosis). In this section, we will analyze the relationship scores for our variables. The method used to calculate this score also varies depending on the type of variable.

4.4 Analyzing the Overall Relationships Among Variables

Categorical variables:

For categorical variables, the score is Cramér's V, which quantifies the strength of association between two categorical variables and it is derived from the chi-square statistic.

Formula for Cramér's V:

$$V = \sqrt{\frac{\chi^2}{n(k-1)}}$$

Where:

χ^2 : Chi-square statistic from the contingency table.

n : Total number of observations.

k : Smaller of number of rows or columns in the contingency table.

Cramer's V ranges from 0 to 1, where:

0: No association between the variables.

1: Perfect association.

It is noteworthy to indicate that Cramer's V does not account for statistical significance but indicates the magnitude of the association. A low score may still be significant if the sample size is large, while a high score may not be significant if the sample size is small.

Numerical variables:

For numerical variables, the score is the Point-Biserial Correlation Coefficient, which measures the strength and direction of the linear relationship between a binary variable (osteoporosis status) and a continuous variable (age).

Formula for Point-Biserial Correlation:

$$r = \frac{\bar{X}_1 - \bar{X}_2}{s} \sqrt{\frac{n_1 n_0}{n^2}}$$

Where:

$\bar{X}_1, \bar{X}_2$: Means of the continuous variable for each group.

s : Standard deviation of the continuous variable.

$n_1 n_0$: Sample sizes for each group.

n : Total sample size.

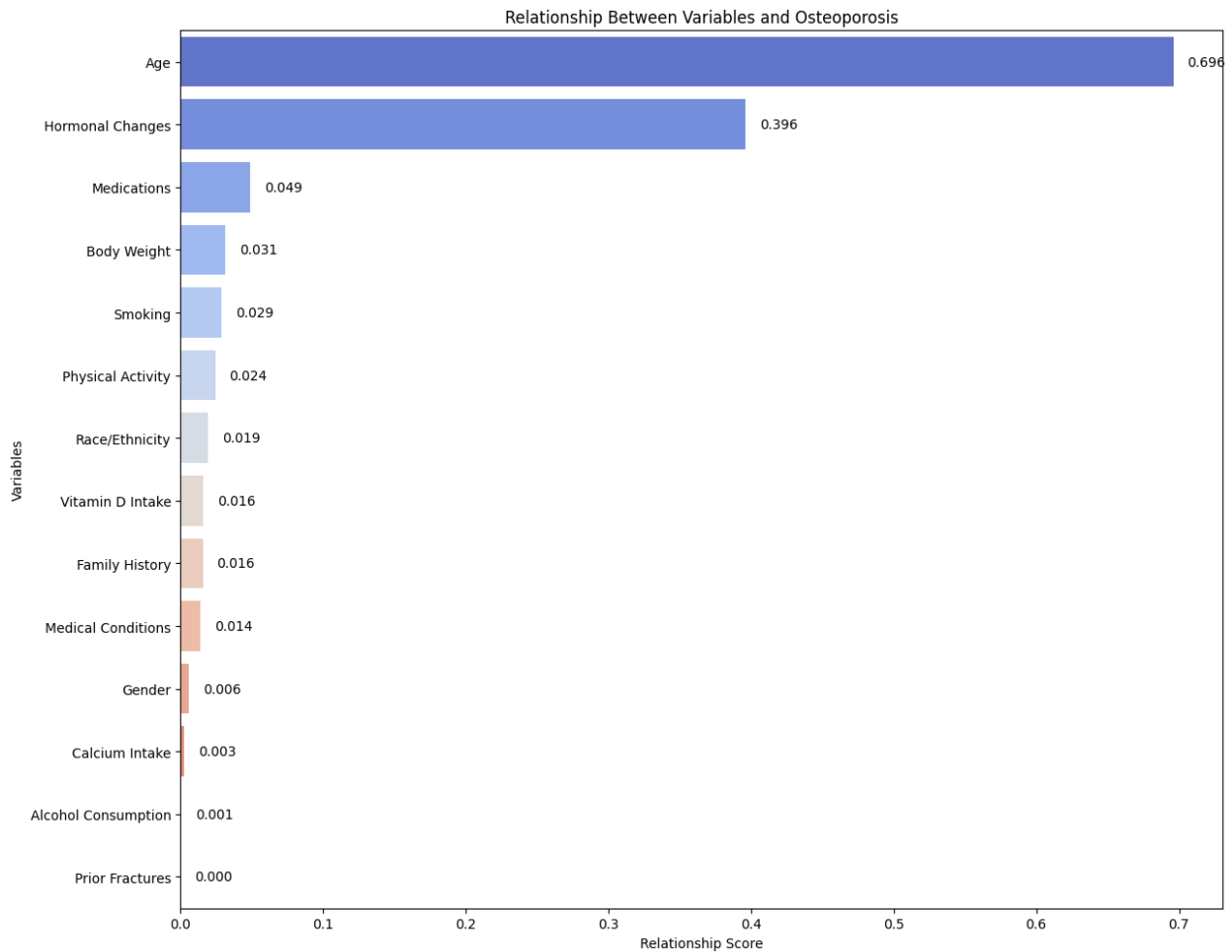
The coefficient ranges from -1 to 1:

-1: Perfect negative correlation.

1: Perfect positive correlation.

0: No linear relationship.

In the next page there is a graph that displays the strength of the relationships between various variables and osteoporosis, represented by the "Relationship Score" on the x-axis. The variables are ranked from the strongest relationship to the weakest, with the tallest bars corresponding to the variables that are most strongly associated with osteoporosis.



Explanation of the Graph:

[Age]

The highest score (~0.70) indicates a strong relationship between age and osteoporosis, suggesting that age is a key factor in determining osteoporosis status.

[Hormonal Changes]

With a score of ~0.40, hormonal changes also show a significant relationship with osteoporosis, consistent with the known biological impact of hormonal changes on bone density.

[Medications]

The relationship between Medications and Osteoporosis is weak, with a score of 0.049. This suggests that medication is not strong enough to be clearly identified in this analysis.

[Body Weight, Smoking]

These variables with score about 0.03 also show weak relationships with Osteoporosis. Their low scores suggest that these factors do not have a very strong correlation with Osteoporosis in your dataset.

[Physical Activity, Race/Ethnicity, Vitamin D Intake, Family History, Medical Condition, Gender]

The relationship between these variables and Osteoporosis is also very weak, with score between 0.01 and 0.02.

[Calcium Intake, Alcohol Consumption]

The relationship between these two variables and Osteoporosis is extremely weak with score lower than 0.01. (Although Calcium Intake is often cited as important factors in Osteoporosis, but in this dataset suggests a very weak association)

[Prior Fractures]

With a zero score, this variable shows virtually no direct association in this dataset.

Heatmap:

The heatmap visually represents the relationships between the variables in the dataset. Since the dataset contains two types of variables—numerical (Age) and categorical (all other variables)—we employ two distinct metrics to evaluate these relationships and combine them into a unified visualization.

Metrics Used:

- ✓ Categorical-Categorical Relationships:

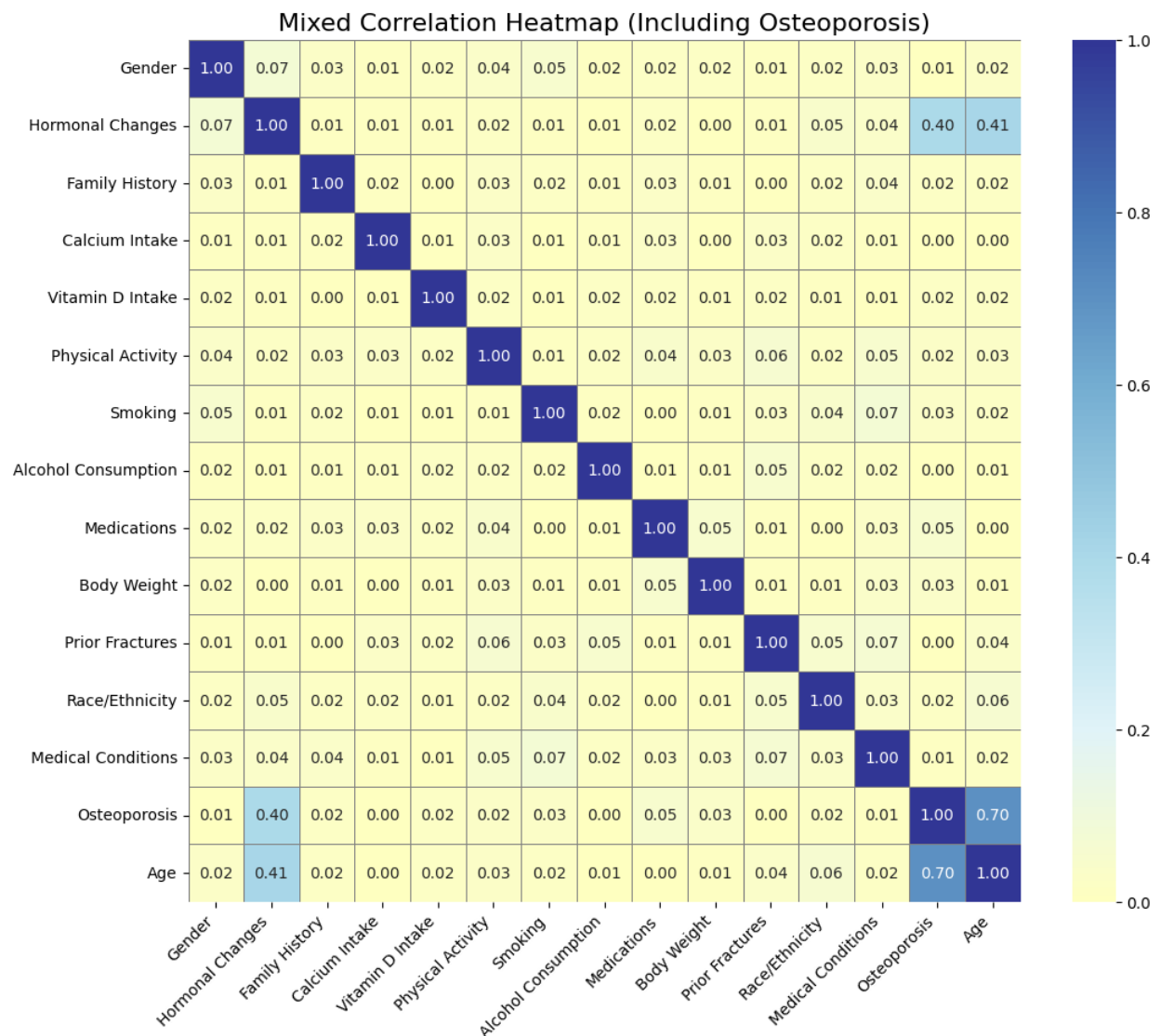
The strength of association between two categorical variables is calculated using Cramér's V. This metric quantifies the association between two categorical variables on a scale from 0 (no association) to 1 (perfect association). It is derived from the chi-square statistic.

- ✓ Categorical-Numerical Relationships:

The relationship between a categorical variable and a numerical variable is measured using the Correlation Ratio (η). This metric evaluates how much the numerical variable's variance is explained by the categorical groups, again on a scale from 0 (no association) to 1 (perfect association).

- ✓ Numerical-Numerical Relationships:

The standard Pearson Correlation Coefficient is used to measure the linear relationship between two numerical variables. This ranges from -1 (perfect negative linear correlation) to +1 (perfect positive linear correlation).



Observations from the Heatmap

[Age and Osteoporosis]

The relationship between Age (numerical) and Osteoporosis (categorical) is strong (0.70), highlighting that osteoporosis is more prevalent as age increases.

[Hormonal Changes and Osteoporosis]

A moderate association (0.40) suggests that hormonal changes play a significant role in predicting osteoporosis.

[Other Variables]

Variables such as Medical Conditions and Prior Fractures have weaker associations with Osteoporosis (close to 0.00), indicating they may not be as predictive or relevant.

5. Model

5.1 Introduction to Modeling

In machine learning, a model is a mathematical representation that captures the relationships between the input features (predictors) and the target variable (outcome). The primary objective of a model is to make accurate predictions on unseen data by learning patterns from the training dataset. This process involves identifying a function $f(X)$ that maps the input features X to the target y , where X represents the predictor variables and y is the outcome of interest.

The modeling process typically consists of the following steps:

1. **Model Selection:**

Choosing the appropriate model based on the nature of the data and the problem (e.g., classification, regression).

2. **Training:**

Using a subset of the data (training set) to optimize the parameters of the model by minimizing a predefined loss function.

3. **Validation:**

Evaluating the model's performance on unseen data (validation set) to tune hyperparameters and prevent overfitting.

4. **Testing:**

Assessing the final performance on a separate test dataset to measure generalization.

For this project, focused on predicting osteoporosis, we explored *supervised learning models*, where the outcome variable (presence or absence of osteoporosis) is known and serves as the basis for training.

We selected the following models for their complementary strengths and relevance to the classification task:

✓ **Logistic Regression:**

A statistical method that models the relationship between the features and the log-odds of the binary outcome.

✓ **Decision Tree Classifier:**

A tree-based model that splits the data into subsets based on feature thresholds, constructing a hierarchy of decisions to classify instances. Decision trees are interpretable and handle non-linear relationships effectively.

The hyperparameters of these models were tuned using Grid Search and evaluated using Cross-Validation to ensure robust performance.

Metrics such as *accuracy*, *precision*, *recall*, and *F1-score* were employed to compare the models, complemented by confusion matrices to gain insight into the nature of classification errors.

In the following sections, we delve deeper into the theoretical foundations and practical applications of each model, explaining how they were implemented, tuned, and evaluated in this project.

5.2 Logistic Regression

Logistic regression, often referred to as the **Logit model**, is a fundamental statistical and machine learning tool for addressing classification problems. It models the relationship between a set of independent variables (features) and a dependent variable (outcome) that is categorical, typically binary. Unlike linear regression, which predicts continuous outcomes, logistic regression predicts the probability that an observation belongs to a particular class, making it well-suited for classification tasks.

The core of the logistic regression model lies in its transformation of the linear relationship between predictors and the response variable through the **logistic function (sigmoid function)**. This function ensures that predicted probabilities are

constrained to the range $[0, 1]$, which aligns with the probabilistic interpretation of classification problems. Mathematically, the logistic function is expressed as:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

where:

$P(Y = 1 | X)$ represents the probability of the positive class.

X_i are the independent variables.

β_i are the coefficients estimated from the data.

and e is the base of the natural logarithm.

The coefficients β are estimated using a method called maximum likelihood estimation (MLE), which finds the parameter values that maximize the likelihood of observing the given data under the model.

Logistic regression is often favored due to its balance of simplicity, interpretability, and effectiveness. While more complex models can offer higher predictive performance in certain scenarios, logistic regression

remains a cornerstone for interpretable and robust classification, particularly when the goal is to understand relationships between variables rather than simply maximize predictive accuracy.

In this chapter, we delve into the logistic regression model, exploring its theoretical foundations, practical implementation, and performance analysis. Additionally, we extend the classical model with regularization techniques such as Ridge (L2) and Lasso (L1) to address overfitting and feature selection, providing a comprehensive framework for handling real-world classification problems.

Our work on Logistic Regression

In our project we applied for the first part Logistic Regression as an inference tool to get more information about predictors and their relation to the predicted variables. To do so we analyzed the dataset three different times each with different assumptions.

For this part of our study we used the built in Logistic Regression from statsmodel since it gives an R-like results which gives a lot more information about the variables and the model itself. The data was already encoded and divided in train and test, so we just have to fit the model and add a constant %beta0.

The first model is applied on the train dataset without any changes.

Logit Regression Results						
Dep. Variable:	Osteoporosis	No. Observations:	1124			
Model:	Logit	Df Residuals:	1109			
Method:	MLE	Df Model:	14			
Date:	Sat, 18 Jan 2025	Pseudo R-squ.:	0.5423			
Time:	10:53:32	Log-Likelihood:	-341.81			
converged:	True	LL-Null:	-746.84			
Covariance Type:	nonrobust	LLR p-value:	7.814e-164			
	coef	std err	z	P> z	[0.025	0.975]
Age	0.1618	0.012	13.917	0.000	0.139	0.185
Gender	0.0419	0.204	0.205	0.838	-0.358	0.442
Hormonal Changes	0.3318	0.272	1.221	0.222	-0.201	0.864
Family History	-0.0124	0.194	-0.064	0.949	-0.393	0.368
Race/Ethnicity	0.0248	0.119	0.208	0.836	-0.209	0.259
Body Weight	0.1412	0.194	0.729	0.466	-0.239	0.521
Calcium Intake	-0.0042	0.194	-0.022	0.983	-0.385	0.376
Vitamin D Intake	0.1118	0.194	0.577	0.564	-0.268	0.491
Physical Activity	-0.0366	0.194	-0.188	0.851	-0.418	0.344
Smoking	-0.2476	0.194	-1.277	0.201	-0.628	0.132
Alcohol Consumption	-0.0465	0.195	-0.238	0.812	-0.429	0.336
Medical Conditions	-0.1138	0.119	-0.955	0.340	-0.347	0.120
Medications	-0.2809	0.198	-1.422	0.155	-0.668	0.106
Prior Fractures	0.2516	0.196	1.282	0.200	-0.133	0.636
const	-5.3418	0.504	-10.595	0.000	-6.330	-4.354

Looking at the results above there's much to discuss. The P-values quantifies how relevant the variables are for the prediction, showing the Age is the most important, followed by Medications, Hormonal Changes, Smoking and Prior Fractures. The "coef" is the weight β of the variables in the regression equation. A positive coef means that the associated variable increase the probability of Y while a negative one decreases it. It does not come as a surprise that "Medications" (since 0 means on medication and 1 the opposite) and "Calcium Intake" have a negative coefficients, but it is unusual to see that "Smoking" has too. As for the positive coefficients the ones that have more importance are Prior Fractures, Hormonal Changes and Vitamin D intake.

To understand how good the model is it's useful to analyze the "Pseudo R-squ", the "Log-Likelihood" and the "LLR p-value". The "Log-Likelihood" shows how good the model fits to the data, the higher it is in respect to the "Log-Likelihood Null", better is the model. Here there is a great difference between the two of them, so the this model is doing a much better job than the null model. The Pseudo R-square is bigger than 0.5, it means that there may be an overfitting problem.

Since "Age" is the most relevant variable but also the only non binary we decided to try a model without it. It is expected to lose a lot of prediction power, but the purpose is to observe if the remaining feature would get more importance.

Logit Regression Results						
=====						
Dep. Variable:	Osteoporosis	No. Observations:	1124			
Model:	Logit	Df Residuals:	1110			
Method:	MLE	Df Model:	13			
Date:	Sun, 19 Jan 2025	Pseudo R-squ.:	0.1323			
Time:	15:45:53	Log-Likelihood:	-648.00			
converged:	True	LL-Null:	-746.84			
Covariance Type:	nonrobust	LLR p-value:	4.091e-35			
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-0.0219	0.255	-0.086	0.932	-0.521	0.477
Gender	0.1990	0.135	1.477	0.140	-0.065	0.463
Hormonal Changes	2.2618	0.196	11.511	0.000	1.877	2.647
Family History	-0.0307	0.134	-0.229	0.819	-0.293	0.232
Race/Ethnicity	-0.0513	0.081	-0.634	0.526	-0.210	0.107
Body Weight	0.1566	0.135	1.161	0.246	-0.108	0.421
Calcium Intake	-0.0766	0.135	-0.569	0.569	-0.341	0.187
Vitamin D Intake	0.0897	0.134	0.669	0.503	-0.173	0.352
Physical Activity	0.0867	0.135	0.643	0.520	-0.178	0.351
Smoking	-0.1563	0.134	-1.162	0.245	-0.420	0.107
Alcohol Consumption	0.0559	0.134	0.416	0.677	-0.207	0.319
Medical Conditions	0.0012	0.082	0.014	0.989	-0.159	0.162
Medications	-0.1934	0.135	-1.432	0.152	-0.458	0.071
Prior Fractures	-0.0091	0.135	-0.067	0.946	-0.273	0.255
=====						

It's pretty clear from the "Log-Likelihood" value, since it's not very different from the null one, that the model doesn't give any useful information since it does not fit well to the dataset.

As for the third and last model we tried manually removing all the variables that have a p-value greater than 0.5 on the first Logit model results.

```
variabili_to_remove2 = ['Family History', 'Race/Ethnicity', 'Gender', 'Alcohol Consumption', 'Calcium Intake', 'Hormonal Changes']
```

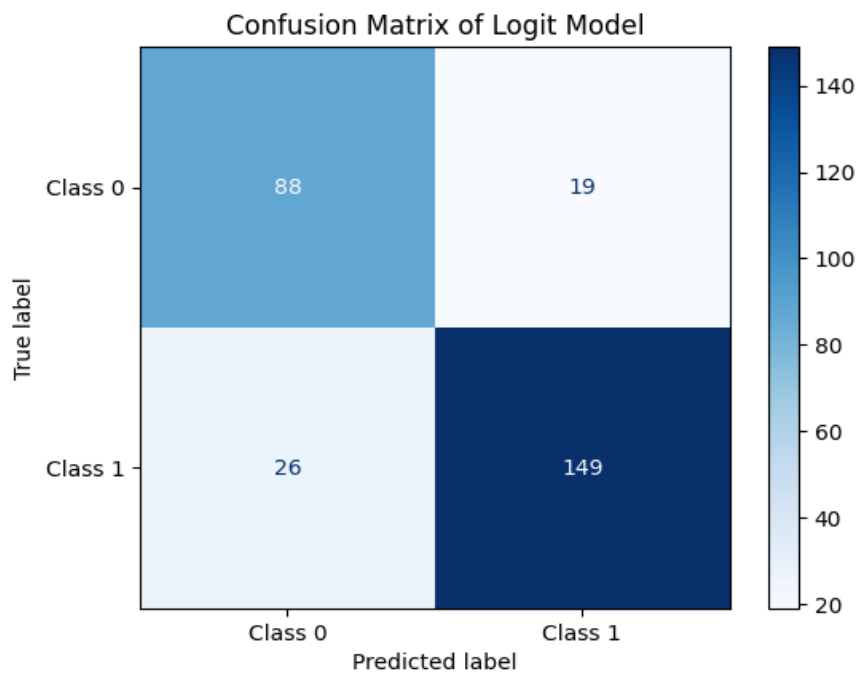
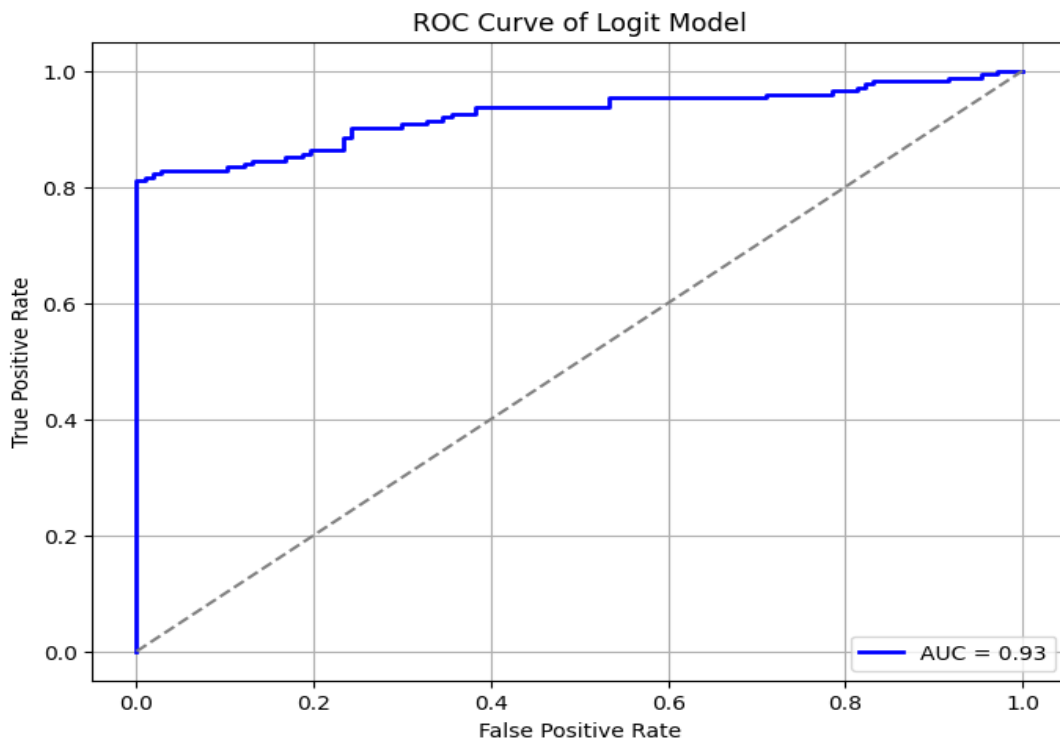
Logit Regression Results						
=====						
Dep. Variable:	Osteoporosis	No. Observations:	1124			
Model:	Logit	Df Residuals:	1115			
Method:	MLE	Df Model:	8			
Date:	Sun, 19 Jan 2025	Pseudo R-squ.:	0.5412			
Time:	15:45:53	Log-Likelihood:	-342.64			
converged:	True	LL-Null:	-746.84			
Covariance Type:	nonrobust	LLR p-value:	3.199e-169			
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-5.4546	0.447	-12.208	0.000	-6.330	-4.579
Age	0.1678	0.011	15.514	0.000	0.147	0.189
Body Weight	0.1473	0.193	0.763	0.445	-0.231	0.526
Vitamin D Intake	0.1157	0.193	0.601	0.548	-0.262	0.493
Physical Activity	-0.0301	0.194	-0.155	0.877	-0.410	0.350
Smoking	-0.2599	0.193	-1.350	0.177	-0.637	0.117
Medical Conditions	-0.1202	0.119	-1.013	0.311	-0.353	0.112
Medications	-0.2607	0.193	-1.350	0.177	-0.639	0.118
Prior Fractures	0.2516	0.194	1.296	0.195	-0.129	0.632
=====						

This model is just slightly worst than the first, showing again how important “Age” variable really is in the “Osteoporosis” prediction on comparison to other variables.

After the preliminary phase where we tried to build the best Logit model, we can apply the trained model on our test sample to see how good it is on prediction checking the results with the help of the ROC Curve (the plot of the true positive rate against the false positive rate) and the Confusion Matrix (to quantify the exact number of good and bad prediction of the model).

Since the second model does not give any information and the third is just a worse version of the first one it would be a waste to do any other analysis on these two, so from now we will just use the first one.



accuracy test	precision test	recall test	f1_score
0.8404255319148937	0.8869047619047619	0.8514285714285714	0.8688046647230321

The AUC score of 0.93 is an excellent score, showing that the model does a pretty good job

As for the Confusion Matrix we added Accuracy test, Precision Test, Recall test and F1 Score too and all of them give show good results since they are all greater than 0.84 .

RIDGE and LASSO Regression

The third Logit model was a failed attempt in getting a better model by manually removing less relevant variables. Now we are going to try if by introducing a penalty in the Loss function with RIDGE regression and LASSO regression we could get a better results.

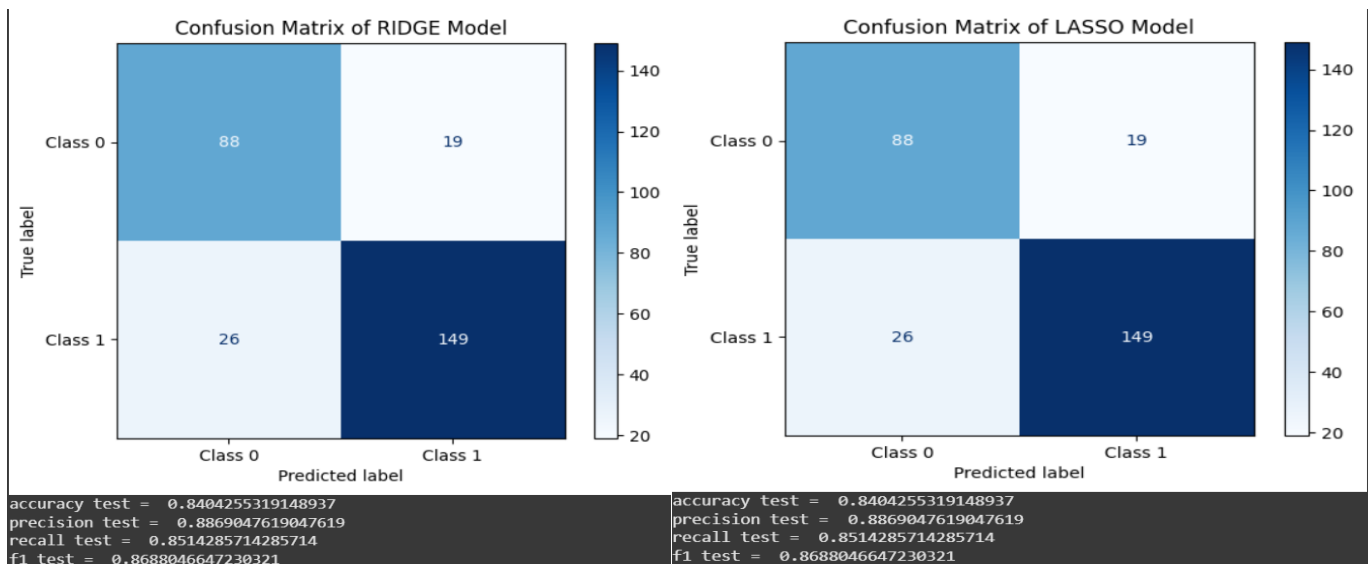
It's harder to apply RIDGE and LASSO regression on the Logistic Regression built in the statsmodel library, so for this second part we are going to use the one from sklearn.

```
from sklearn.linear_model import LogisticRegression
```

```
#Ridge Logistic Regression on train dataset
ridge_model = LogisticRegression(penalty='l2', solver='liblinear', C=1)
ridge_model.fit(x_logit, y_train_fold)
```

```
# Modello Ridge Logistic Regression
lasso_model = LogisticRegression(penalty='l1', solver='liblinear', C=1.0)
lasso_model.fit(x_logit, y_train_fold)
```

After the training process we then apply both model on the test set and, just like we did for the original Logistic regression, we obtain the Confusion Matrix with all the other test (accuracy, precision ,...).



As shown by these results RIDGE and LASSO do not make any changes in the prediction, proving that there is no need for any kind of regularization. This was to be expected since there doesn't seem to be any multicollinearity between features nor overfitting in the model.

5.3 Decision Tree

The Decision Tree is a supervised machine learning method particularly suited for both classification and regression tasks. It is a non-parametric, distribution-free approach, widely applied across various domains such as medicine, genetics, finance, and marketing. In addition to its predictive capabilities, decision trees serve as an intuitive tool for exploring and understanding data, thanks to their clear and easily interpretable graphical structure.

A decision tree works by recursively partitioning the dataset into increasingly simple and homogeneous subsets with respect to the target variable. At each node, a "question" is posed—a splitting rule based on the predictors—that divides the data into two groups. This process continues until reaching the terminal nodes, or leaves, where a specific value or class of the target variable is assigned.

One of the key aspects of decision trees is the criterion used to determine the splits, or the questions that best divide the data. These decisions are guided by impurity measures, such as the Gini index or entropy (the formulas for the former and the latter are provided below), which assess how well a split improves the homogeneity of the resulting groups with respect to the target variable. This approach enables the model to identify the most relevant predictors while automatically discarding less significant ones.

$$G = 1 - \sum_{i=1}^n p_i^2$$

Where:

- G: Gini Index
- n: Number of classes
- p_i : Proportion of observations belonging to class i

$$H = - \sum_{i=1}^n p_i \log_2(p_i)$$

Where:

- H: Entropy
- n: Number of classes
- p_i : Proportion of observations belonging to class i

Decision trees offer several advantages. First, they are highly interpretable and they can handle both categorical and numerical data seamlessly, making them versatile across different types of datasets. Moreover, decision trees perform automatic feature selection, focusing on the most important predictors while disregarding irrelevant ones.

In our project, the decision tree proved to be an ideal tool for analyzing the factors influencing the onset of osteoporosis. Its ability to visually and intuitively represent the relationships between predictors and the target variable makes it particularly valuable in medical research. Furthermore, the rule-based decision-making process

provides clear and actionable insights, facilitating practical applications and communication of results.

5.3.1 Description of the Decision Tree Optimization Process

In this project, we implemented a structured process to optimize and evaluate the Decision Tree Classifier using a combination of Grid Search and Cross-Validation. The main goal was to identify the hyperparameter configuration that maximizes the model's performance in terms of accuracy, while ensuring robustness and generalizability.

Defining the Parameter Grid:

We defined a parameter grid containing some of the most relevant settings for a decision tree. These include:

```
# Define the parameter grid for the Decision Tree
param_grid = {
    'criterion': ['gini', 'entropy'],
    'max_depth': [3, 5, 7, 10], # Limit tree depth
    'min_samples_split': [5, 10, 20], # Ensure splits have sufficient data
    'min_samples_leaf': [5, 10, 20], # Ensure leaves have sufficient data
    'max_features': [None, 'sqrt', 'log2'] # Regularize feature usage
}
```

Creating the Classifier and Stratified K-Fold Cross-Validation:

To ensure reliable results, we used a Stratified K-Fold Cross-Validation strategy with 5 folds, this tool is available in the library scikit-learn. This approach splits the data into five subsets while maintaining the proportional distribution of classes, thus avoiding biases in the results. This technique is particularly useful for classification problems where imbalanced classes can distort model performance.

We then initialized the tree Classifier with a fixed `random\_state`, ensuring reproducibility of the training process across different runs. Since our target variable (i.e. whether a diagnosis of osteoporosis is present or not) is moderately imbalanced, the parameter **class_weight = 'balanced'** was added.

The class_weight equals to 'balanced' option automatically adjusts the weights assigned to each class inversely proportional to their frequencies in the dataset. This ensures that the model pays equal attention to both the minority and majority classes.

The weights are calculated using the formula:

$$w_c = \frac{n_{samples}}{n_{classes}n_c}$$

Where:

- w_c : Weight for class c
- $n_{samples}$: Total number of samples in the dataset
- $n_{classes}$: Total number of classes
- n_c : Number of samples in class c

```
# Initialize the Decision Tree Classifier
dt_classifier = DecisionTreeClassifier(class_weight='balanced', random_state=42)
```

Optimization with Grid Search:

To find the best parameter combination, we used Grid Search from the library `scikit_learn`, which systematically evaluates each combination in the grid by training and validating the model on each fold of the Cross-Validation. This process not only identifies the configuration that maximizes the average accuracy but also provides a measure of performance variability through the standard deviation calculated across the folds.

```
# Configure GridSearchCV with Cross-Validation
grid_search = GridSearchCV(
    estimator=dt_classifier,
    param_grid=param_grid,
    scoring='recall',
    cv=skf,
    n_jobs=-1 # Use all CPU cores for parallel computation
)
```

Grid Search was configured to use all CPU cores, optimizing computational efficiency. Recall was chosen as the primary metric to evaluate model performance, as it is particularly important for medical datasets. Prioritizing recall helps minimize the number of false negatives, which are the most dangerous outcomes in the medical domain.

```
Best Parameters: {'criterion': 'entropy', 'max_depth': 10, 'max_features': None, 'min_samples_leaf': 5, 'min_samples_split': 5}
Best Cross-Validation Score: 0.8875073891625614
Standard Deviation: 0.01823209254687563
Test Accuracy: 0.9359886201991465
```

The model was optimized using hyperparameter tuning, selecting parameters such as the 'entropy' criterion for splits, a maximum tree depth of 10, no restriction on the number of features for splits, and minimum sample requirements of 5 for both leaf nodes and node splits. The best cross-validation score achieved was 0.8875, with a standard deviation of 0.0182 calculated across the folds of the cross-validation, meaning it reflects the variation in performance between the different folds. When tested on the independent dataset, the model achieved an accuracy of 93.59%, demonstrating strong generalization ability.

5.3.2 Evaluation, and Reporting Process:

The **Decision Tree Classifier** is explicitly initialized with the hyperparameters that represent the optimal values identified during the previous Grid Search phase, designed to maximize the model's performance.

The classifier is trained using the fit method (**.fit()**) applied to the training data of the current fold. This allows the model to build a tree structure that minimizes entropy at each node, learning the patterns present in the data. The dataset was split into training and test sets using the `train_test_split` function from the `sklearn.model_selection` library. This ensures that 20% of the data is reserved for testing, with stratification applied to preserve the proportion of each class in both the training and test sets.

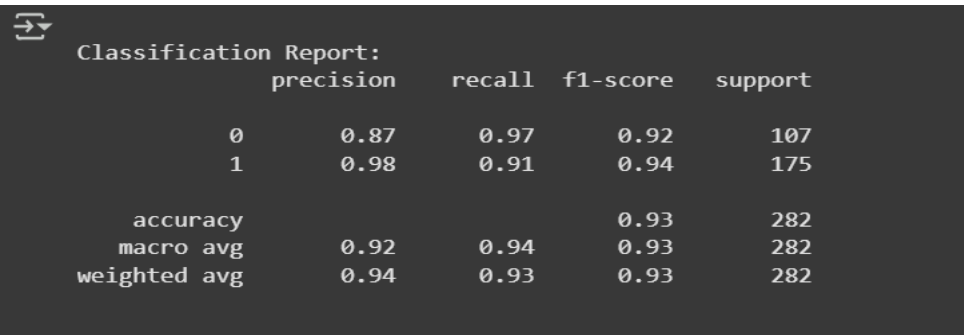
The `stratify = y` parameter in the `train_test_split` function is used to ensure that the split between the training and test sets maintains the same proportion of classes as in the original dataset.

After training, the model is used to predict the labels of the test fold using the `predict` method (**.predict()**). The predictions (`'y_pred_fold'`) are then compared with the actual labels of the test set (`'y_test_fold'`) to evaluate performance.

The model's performance is analyzed using :

1. Classification Report:

Provides precision, recall, F1-score, and support for each class, offering a detailed overview of the performance (see Appendix for the metrics' formula).



```
Classification Report:
      precision    recall  f1-score   support

     0       0.87      0.97      0.92        107
     1       0.98      0.91      0.94        175

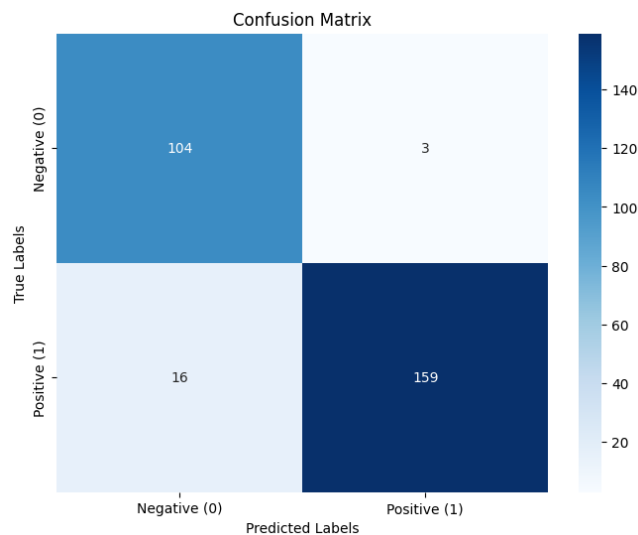
 accuracy      0.93        282
 macro avg      0.92        282
 weighted avg      0.94        282
```

Classification Report:				
	precision	recall	f1-score	support
0	0.87	0.97	0.92	107
1	0.98	0.91	0.94	175
accuracy			0.93	282
macro avg	0.92	0.94	0.93	282
weighted avg	0.94	0.93	0.93	282

The model demonstrates strong overall performance, achieving an **accuracy of 93%**. For class **0**, the model is better at identifying true negatives (recall of 97%), while for class **1**, it is better at avoiding false positives (precision of 98%). The weighted averages for precision, recall, and F1-score are all above 93%, indicating balanced and consistent performance across both classes.

2. Confusion Matrix:

Summarizes correct and incorrect predictions in a tabular format, distinguishing true positives, true negatives, false positives, and false negatives.

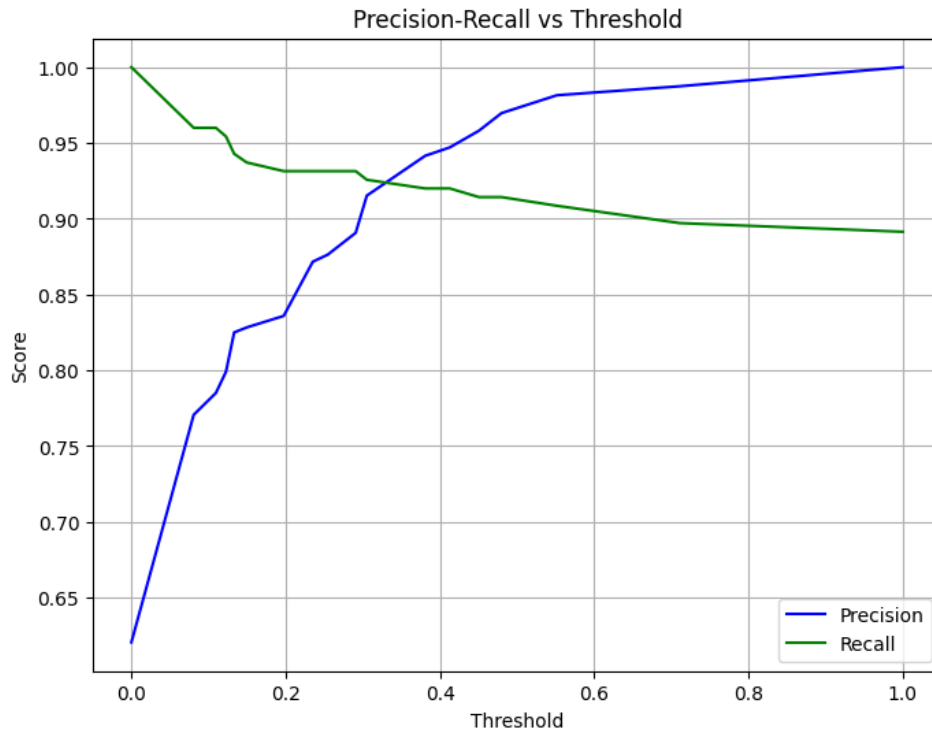


So the false negatives account for 5.67%.

Test Accuracy:	0.9326
Train Accuracy:	0.9297

The very small difference between the training and test accuracy suggests that the model generalizes well and is not overfitting to the training data. Overfitting typically occurs when the model performs significantly better on the training set than on the test set, which is not the case here.

To further improve the model and address the issue of false negatives, we attempted to optimize the trade-off between precision and recall by adjusting the decision threshold. False negatives are particularly critical in this context, and reducing their occurrence can enhance the model's effectiveness, especially for applications where missing positive cases is highly undesirable.



The plot, "Precision-Recall vs Threshold," was generated to guide the selection of an optimal threshold. After analyzing the plot, we set the threshold to 0.4, which offers a reasonable balance between precision and recall.

With the addition of the threshold adjustment, the model achieves slightly improved results, as demonstrated by the updated confusion matrix and classification report.

Classification Report:					
	precision	recall	f1-score	support	
0	0.88	0.92	0.89	107	
1	0.95	0.92	0.93	175	
accuracy			0.92	282	
macro avg	0.91	0.92	0.91	282	
weighted avg	0.92	0.92	0.92	282	

The new confusion matrix is:

$$\begin{bmatrix} 98 & 9 \\ 14 & 161 \end{bmatrix}$$

With a false negative percentage of 4.96%, which is slightly lower compared to the model without the threshold.

Overall, the accuracy remains high at 92%, with macro and weighted averages for precision, recall, and F1-score consistently above 91%. These results indicate that the threshold adjustment has helped refine the model's sensitivity to positive cases, reducing false negatives while maintaining balanced performance across both classes.

3. Conclusion Decision Tree:

The decision tree model provides a comprehensive understanding of the factors influencing the target variable. From the decision tree visualization, it is evident that several features, such as hormonal changes, family history, smoking, and physical activity, play a role in classifying outcomes. However, the feature importance analysis reveals that age overwhelmingly dominates as the most significant predictor, contributing over 90% to the model's decision-making process. This highlights that age alone carries substantial predictive power in the dataset, overshadowing the influence of all other features.

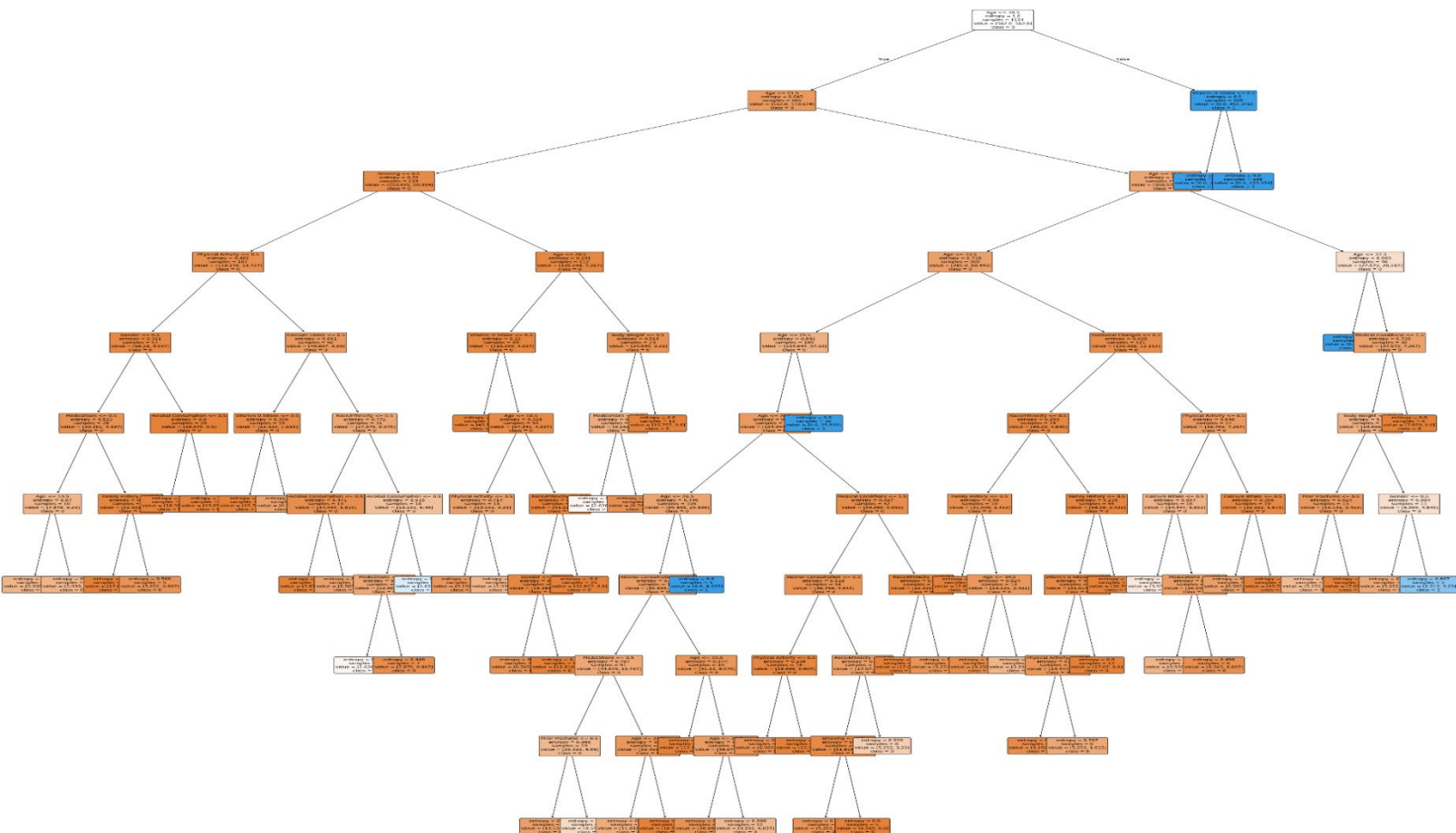
The low importance values for other features, such as family history, hormonal changes, smoking, and physical activity, indicate that while they are used in the tree for specific splits, their overall contribution to reducing impurity is minimal compared to age.

The decision tree provides a visual representation of how the dataset is split to classify the target variable. Starting from the root node at the top, the tree divides the data based on a feature (the age in our case) and a threshold condition ("Age < 38,5").

Each node in the tree shows:

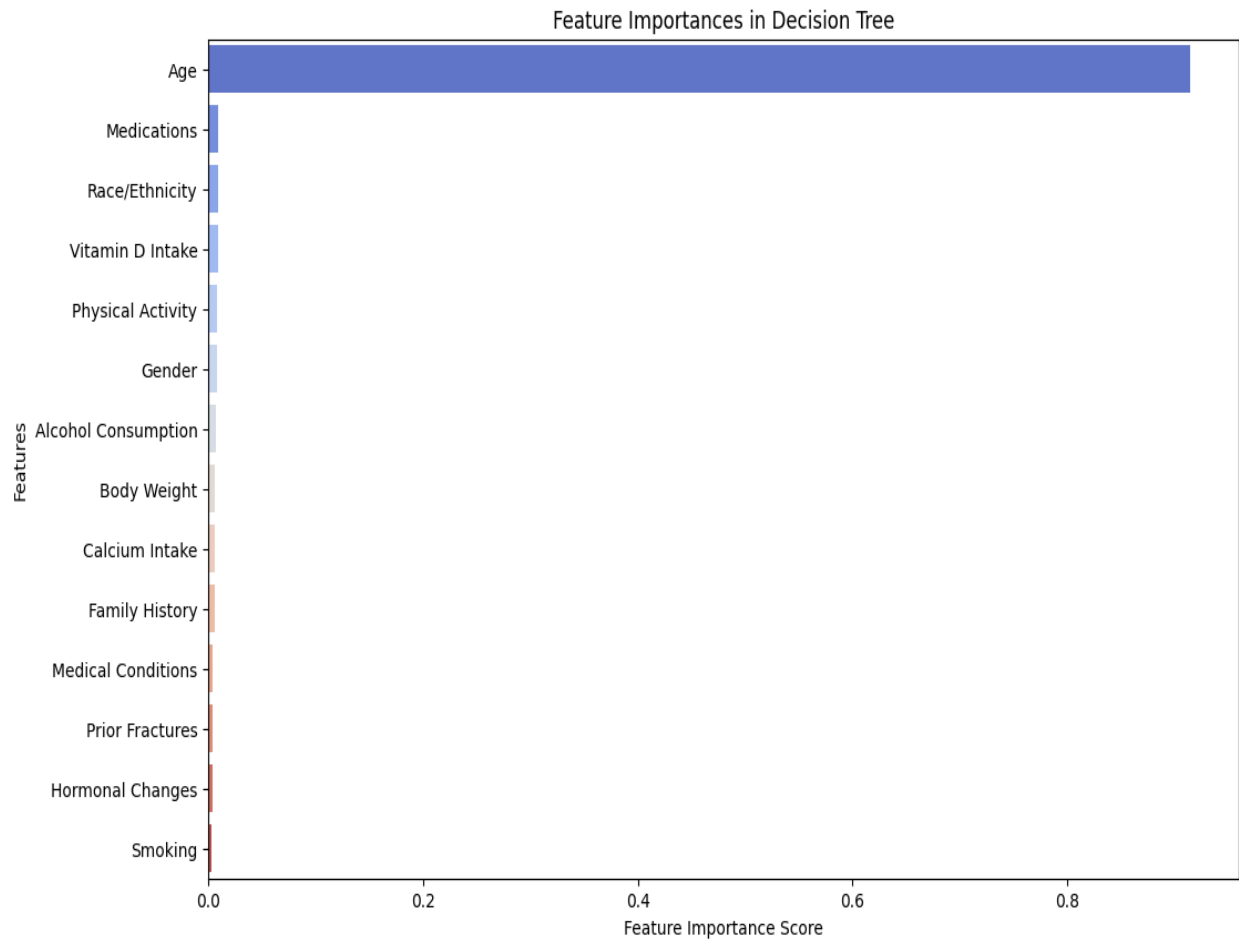
- The feature and threshold used for the split.
- The number of samples reaching that node.
- The class distribution of the samples in that node.

The tree continues to split the data until it reaches leaf nodes, which represent the final predictions. A leaf node displays the predicted class along with the distribution of samples for each class. The color of the nodes indicates their purity, with darker shades representing nodes where most samples belong to a single class.



Feature Importance:

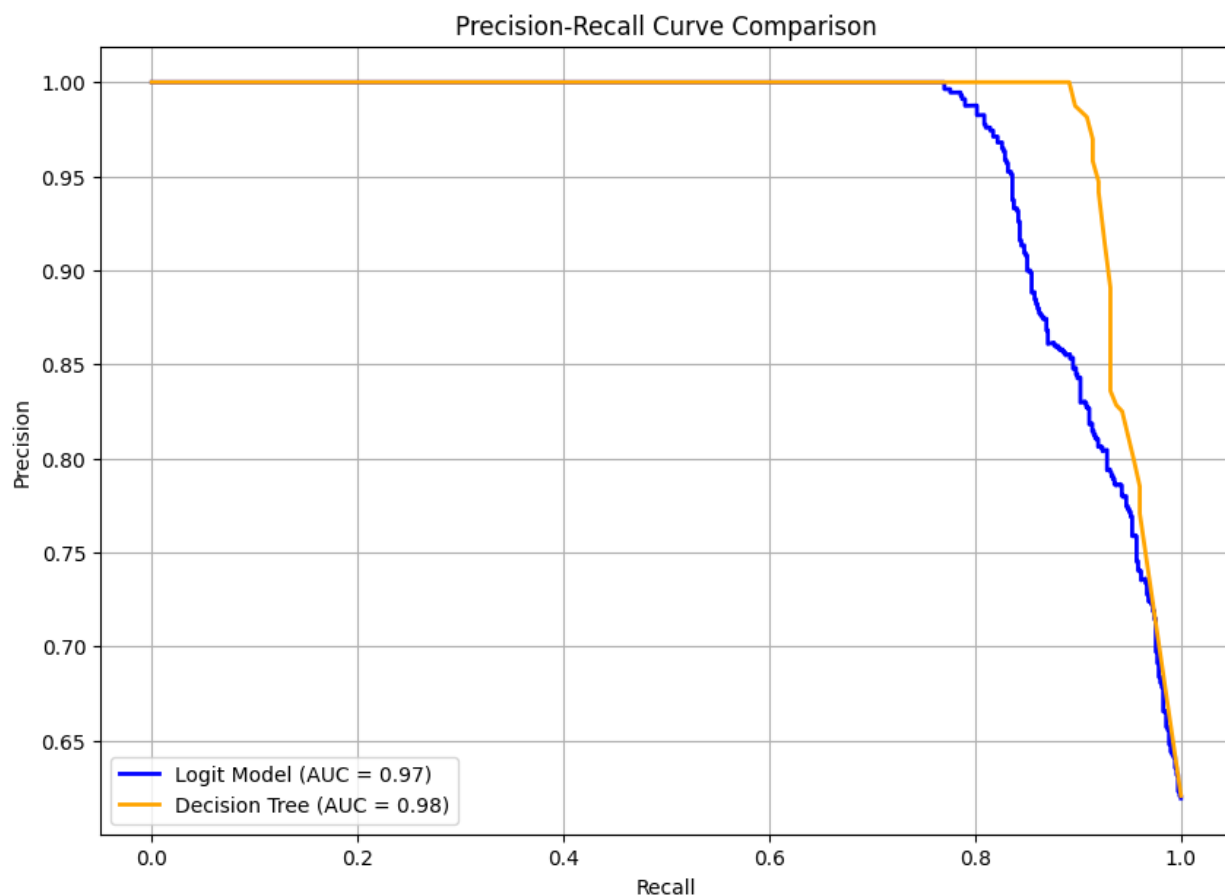
The bellow graph shows the Feature Importance in the Decision tree



5.4. Comparison Between Logistic and Decision Tree

Precision-Recall Curve Analysis

The Precision-Recall curves above compare the performance of two models: Logit Model (Logistic Regression) and Decision Tree.



Decision Tree:

The Decision Tree achieved an AUC of 0.98, reflecting exceptional performance. Its curve remains close to the top-right corner (Precision ≈ 1 , Recall ≈ 1) for a large portion of the graph, indicating high precision while maintaining excellent recall.

Such strong performance could raise concerns about overfitting. However, with Train Accuracy (92.97%) and Test Accuracy (93.26%) being nearly identical, this suggests that the Decision Tree generalizes well and is not overfitting. The balanced accuracy values confirm its robustness.

Logit Model:

The Logit Model performed surprisingly well, achieving an AUC of 0.97, which is very close to the Decision Tree. The curve shows that the model maintains high

precision across a significant range of recall, indicating its ability to capture the underlying patterns in the data effectively.

This improvement compared to earlier results suggests that the model's configuration and the dataset's characteristics allow it to perform nearly on par with the Decision Tree, even though it is a simpler, linear model.

Dataset Characteristics:

The high AUC values for both models can likely be attributed to the artificial nature of the dataset, which provides clear patterns that both models can leverage effectively. In real-world datasets, with more noise and complexity, the performance gap between the two models might become more pronounced.

Both the Decision Tree and Logit Model show strong performance on this dataset, with AUC values of 0.98 and 0.97, respectively.

While the Decision Tree slightly outperforms the Logit Model, the latter's strong results indicate that simpler, interpretable models can also perform exceptionally well in scenarios where the dataset is well-structured and artificial in nature.

6.Conclusions

In our study on osteoporosis, we identified that age emerges as the most significant factor associated with the condition, to the extent that it overshadows the influence of other variables. This result was consistently observed across all stages of the analysis, including Exploratory Data Analysis (EDA), logistic regression, and the decision tree. Age, already well-documented in the literature as a key determinant of bone density loss, is reaffirmed here as a critical factor in our dataset.

Secondary factors that also showed a significant contribution include:

1. Hormonal changes, such as those associated with menopause, which represent a critical biological transition that accelerates bone density loss.
2. Corticosteroid use, known to adversely affect bone metabolism and increase the risk of osteoporosis.

These findings are strongly supported by the scientific literature:

- Age has been identified as a major risk factor for osteoporosis, as described by Prof John A Kanis MD "Diagnosis of osteoporosis and assessment of fracture risk", highlighting the decline in bone mineral density (BMD) and the increased risk of fractures with advanced age.³
- The role of hormonal changes, particularly estrogen depletion during menopause, in bone health has been well-documented by Cannarella R, Barbagallo F, et al. in "Osteoporosis from an Endocrine Perspective: The Role of Hormonal Changes in the Elderly". J Clin Med. 2019 Oct 1;8(10):1564. doi: 10.3390/jcm8101564. PMID: 31581477; PMCID: PMC6832998.. (2002)**.⁴
- The detrimental effects of corticosteroids on bone density have been highlighted by Mitra R. " Adverse effects of corticosteroids on bone metabolism: a review." (PM R. 2011 May;3(5):466-71; quiz 471. doi: 10.1016/j.pmrj.2011.02.017. PMID: 21570035.), emphasizing their contribution to bone fragility.⁵

³ <https://www.sciencedirect.com/science/article/abs/pii/S0140673602087615>

⁴ <https://pmc.ncbi.nlm.nih.gov/articles/PMC6832998/>

⁵ <https://www.sciencedirect.com/science/article/abs/pii/S0140673602087615>

The combined use of statistical methods and machine learning has strengthened the validity of our findings. Logistic regression quantified the individual contribution of each variable, while the decision tree visually highlighted the dominant role of age in predictive decisions.

But Don't Worry About Your Habits!

Interestingly, some of the variables we might think are important—such as smoking, alcohol consumption, and vitamin intake—barely contributed to the predictions of osteoporosis in our analysis. So, at least when it comes to osteoporosis, you can rest easy: skipping your vitamins or indulging in a smoke might not be the end of the world (for your bones, at least!). However, it's always good to keep in mind their broader health effects!

Implications

These results underline the need for targeted prevention strategies, particularly focused on older populations, and prudent management of corticosteroid use in at-risk patients. Additionally, interventions aimed at monitoring hormonal changes could provide further benefits in managing osteoporosis.

Appendix

Explanation of Metrics:

1. Precision

- Definition: Precision measures the proportion of true positive predictions out of all positive predictions made by the model. It indicates how accurate the model is when it predicts a positive label.
- Formula:
$$Precision = \frac{True\ Positives\ (TP)}{True\ positives\ (TP) + False\ positives\ (FP)}$$

2. Recall

- Definition: Recall measures the proportion of actual positives correctly identified by the model. It evaluates the model's ability to capture all positive cases.
- Formula:
$$Recall = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + False\ negatives\ (FN)}$$

3. F1-Score

- Definition: The F1-score is the harmonic mean of precision and recall. It provides a single measure of performance by balancing precision and recall, especially when the dataset is imbalanced.
- Formula:
$$F1 = 2 \frac{Precision \cdot Recall}{Precision + Recall}$$

4. Support

- Definition: Support refers to the number of actual instances of each class in the test dataset.

5. Accuracy

- Definition: Accuracy is the proportion of correctly classified samples out of the total number of samples.
- Formula:
$$Accuracy = \frac{True\ Positives\ (TP) + True\ Negatives\ (TN)}{Total\ Samples}$$

6. Macro Average

- Definition: The macro average computes the unweighted mean of precision, recall, and F1-score across all classes, treating each class equally.

7. Weighted Average

- Definition: The weighted average takes into account the number of instances (support) for each class when calculating the mean precision, recall, and F1-score.

Explanation of Regularization Techniques

1. Ridge Penalty (L2 Regularization)

Definition:

Ridge penalty, also known as L2 regularization, adds a penalty term proportional to the square of the magnitude of coefficients to the loss function. It discourages large coefficients and is particularly useful for addressing multicollinearity in linear regression models.

Impact:

By shrinking coefficients, Ridge regression reduces model variance and improves generalization. However, it does not perform feature selection, as it keeps all coefficients but shrinks them closer to zero.

2. Lasso Penalty (L1 Regularization)

Definition:

Lasso penalty, or L1 regularization, adds a penalty term proportional to the absolute value of coefficients to the loss function. It encourages sparsity in the model by shrinking some coefficients to exactly zero, effectively performing features selection.

Impact:

Lasso regression is particularly effective when the dataset has many irrelevant features, as it can automatically exclude them by setting their coefficients to zero. However, it can sometimes over-penalize and ignore important correlated features.